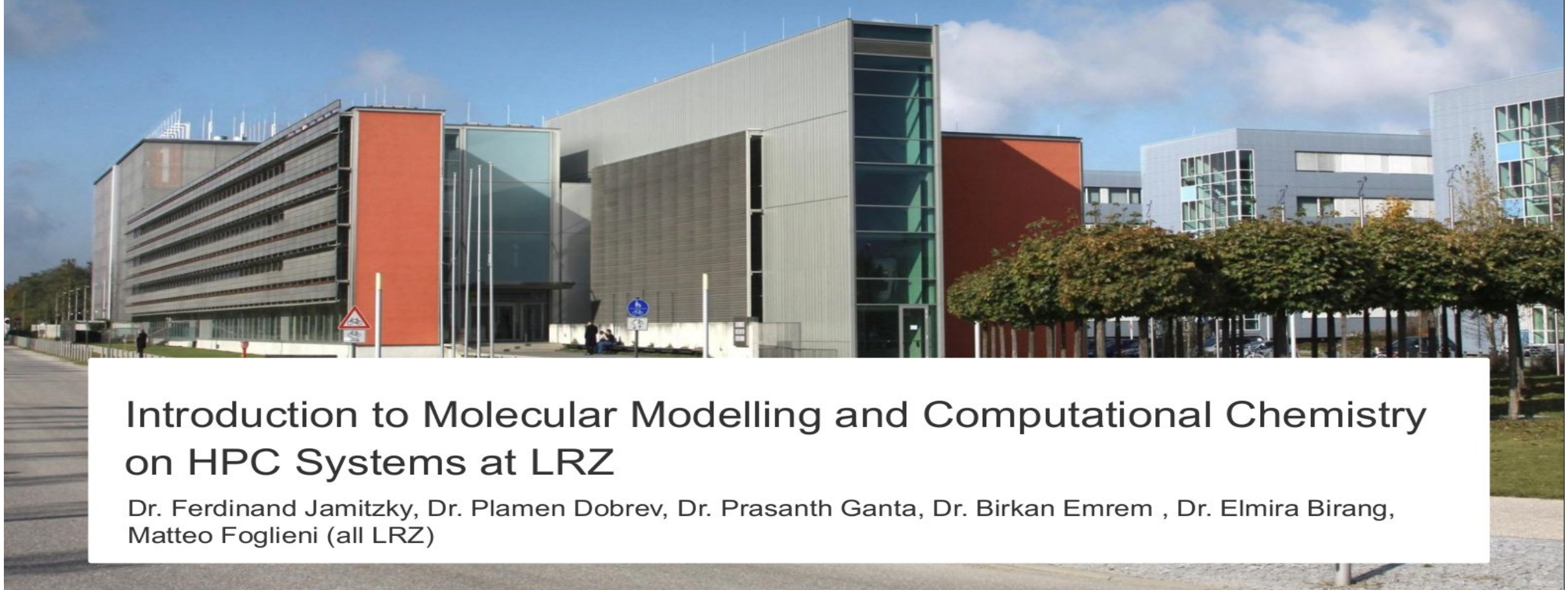


Leibniz-Rechenzentrum
der Bayerischen Akademie der Wissenschaften

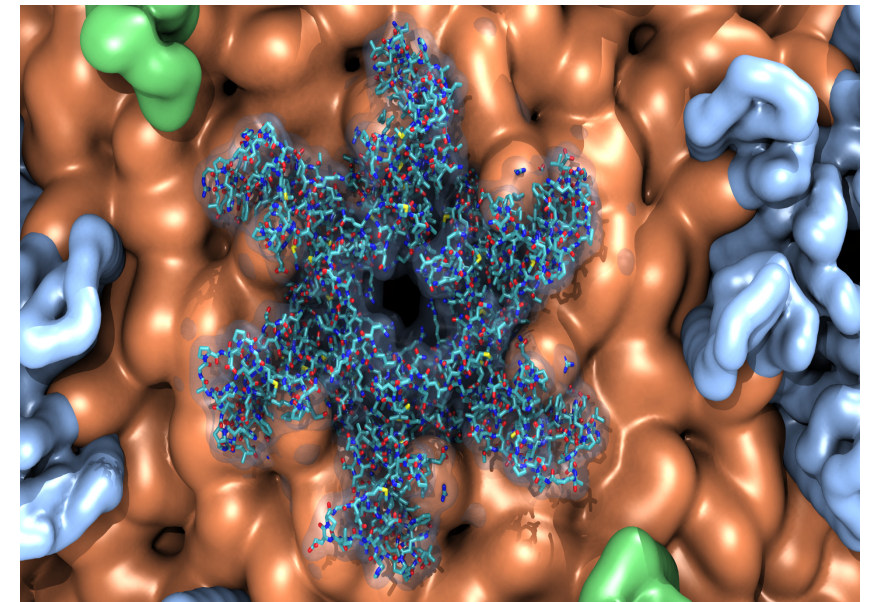
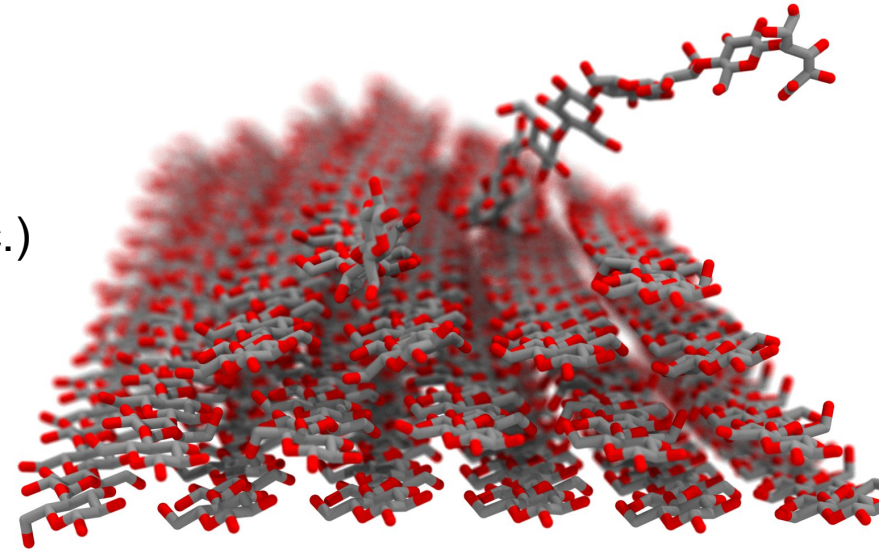
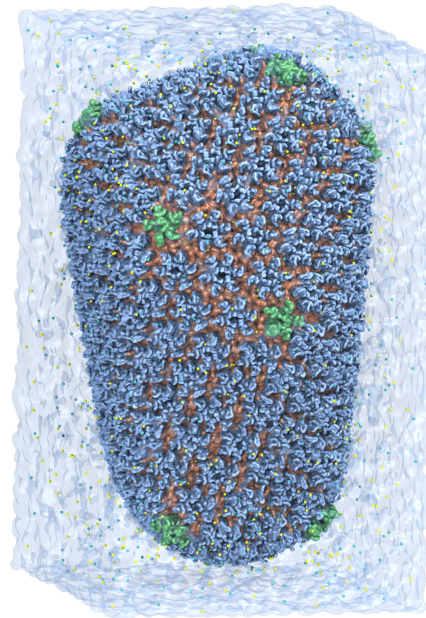


Introduction to Molecular Modelling and Computational Chemistry on HPC Systems at LRZ

Dr. Ferdinand Jamitzky, Dr. Plamen Dobrev, Dr. Prasanth Ganta, Dr. Birkan Emrem , Dr. Elmira Birang,
Matteo Foglieni (all LRZ)

Introduction to VMD (Visual Molecular Dynamics)

- Visualizing large Molecular Dynamics Trajectories
- Supports many molecular formats (PDB, PSF, DCD, etc.)
- Commonly used with NAMD and CHARMM
- Open-source and cross-platform
- Data Analysis tools
- HD images and Movie making
- Built in tcl and python scripting



Installing VMD on Linux OS

1. Go to Download: www.ks.uiuc.edu/Research/vmd

Version 2.0.0 LATEST ALPHA (2025-03-13) Platforms:

Latest pre-release ALPHA test version

- **LINUX_64 (RHEL 8+)** OpenGL, CUDA, OptiX RTX, RTX RTX

Version 1.9.4 LATEST ALPHA (2023-06-08) Platforms:

Latest pre-release ALPHA test version

- **Source Code**
- **LINUX_64 (RHEL 7+)** OpenGL, CUDA, OptiX RTX, OSPRay
- **LINUX_64 (RHEL 7+)** OpenGL, CUDA, OptiX RTX, OSPRay
- **MacOS 11.x, ARM64 (64-bit "M1" Macs)** (Apple MacOS-X 11)
- **MacOS 10.15, x86_64 (64-bit Intel x86_64)** (Apple MacOS-X)
- **Windows 64-bit, CUDA, OptiX, OSPRay (64-bit Intel x86_64)**



2. Register

Registration/Login

You will need a username and password to download software.

If this is your first download, please choose a username and password to register.
Current NAMD or VMD users, please enter your existing username and password.

Username:

Password:

Your download will continue after you have registered or logged in.

3. VMD folder contents

```
di97gef@badwlrz-cl18014:vmd-1.9.4a57$ ls
Announcement      distrib      msvc        scripts
bin               doc          plugins     shaders
configure         lib          proteins    src
configure.options LICENSE      python
data              LINUXAMD64  README
```

4. modify configure file (optional)

```
#####
# User modifiable installation parameters, can be overridden by env variables
#####
# Name of shell script used to start program; this is the name used by users
$install_name = "vmd";

# Directory where VMD startup script is installed, should be in users' paths.
$install_bin_dir="/home/di97gef/.local/bin/";

# Directory where VMD files and executables are installed
$install_library_dir="/home/di97gef/Documents/packages/vmd/";
```

Installing VMD on Linux OS

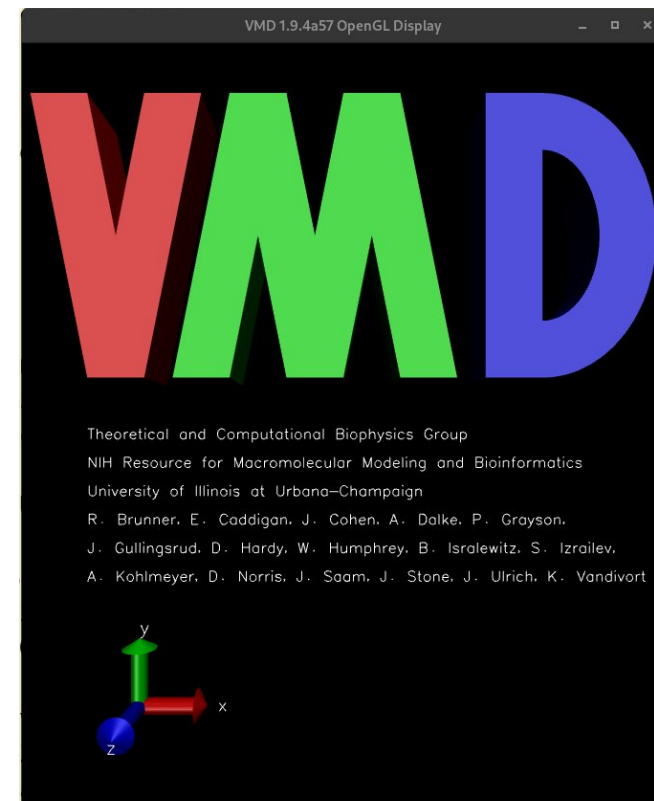
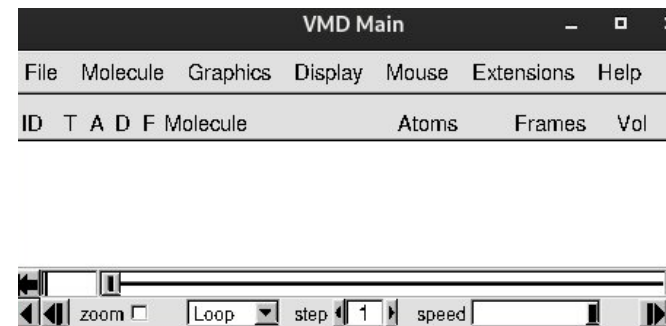
```
# Architectures:
# -----
# AIX6_64      - compile on an RS/6000 machine (gcc/g++/xlc/xlc)
# ANDROIDARMV7A - cross-compile for Android on ARM V7A
# BLUEWATERS   - compile for NSF/NCSA Blue Waters
# CRAY_XC      - compile for Cray XCx0 machines with GPUs
# CRAY_XK      - compile for Cray XKx machines with GPUs
# FREEBSD      - compile on a FreeBSD machine (gcc/g++)
# FREEBSDAMD64 - compile on a FreeBSD machine (gcc/g++)
# LINUX        - compile on a Linux machine (x86 32-bit)
# LINUXAMD64   - compile on a Linux machine (x86 64-bit AMD64 or EM64T)
# LINUXARM64   - compile on a Linux machine (64-bit ARM)
# LINUXPPC64   - compile on a Linux machine (64-bit PowerPC)
# MACOSX       - compile on a MacOS-X PowerPC machine
# MACOSXARM64  - compile on a MacOS-X ARM64 machine, 64-bit ABI
# MACOSXX86    - compile on a MacOS-X x86 machine, 32-bit ABI
# MACOSXX86_64 - compile on a MacOS-X x86 machine, 64-bit ABI
```

5. Run configure

```
di97gef@badwlrz-cl18014:vmd-1.9.4a57$ ./configure LINUXAMD64
```

6. Go to src folder and vmd install

```
di97gef@badwlrz-cl18014:src$ make install
if [ ! -d "/home/di97gef/Documents/packages/vmd/" ]; then \
    mkdir -p "/home/di97gef/Documents/packages/vmd/" ; \
fi ; \
if [ ! -d "/home/di97gef/.local/bin/" ]; then \
    mkdir -p "/home/di97gef/.local/bin/" ; \
fi ; \
if [ ! -d "/home/di97gef/Documents/packages/vmd"/doc ]; then \
    mkdir -p "/home/di97gef/Documents/packages/vmd"/doc ; \
fi
```



VMD in this course

<https://vm-138-246-237-46.cloud.mwn.de:8443/>

 **Leibniz Supercomputing Centre**
of the Bavarian Academy of Sciences and Humanities

integrated, single access point for all of your HPC applications

Desktop Apps


Remote Desktop
Container

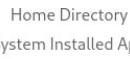
Development


Code Server for VS Code


Jupyter Notebook

Files


Home Directory


System Installed App

Remote Desktop Container

This application uses noVNC to connect to a remote desktop session hosted within a container on a compute node, provided by a VNC server.

If a connection session has timed out, the button "Launch Remote Desktop" can be used to re-enter the session. Because of security concerns, attempting to rejoin directly from noVNC will not work.

Select a container image:

Rocky 9.3 | XFCE4 | Charliecloud | SquashFS

Select a cluster partition for your job: *node-small*

cpu@red-lion::node

Number of CPU nodes

1

Number of cores per job task on a single node - CPU flavor

Medium 8 cores

The real memory required per node - RAM flavor

Small 32 GB

Total run time in hours

1

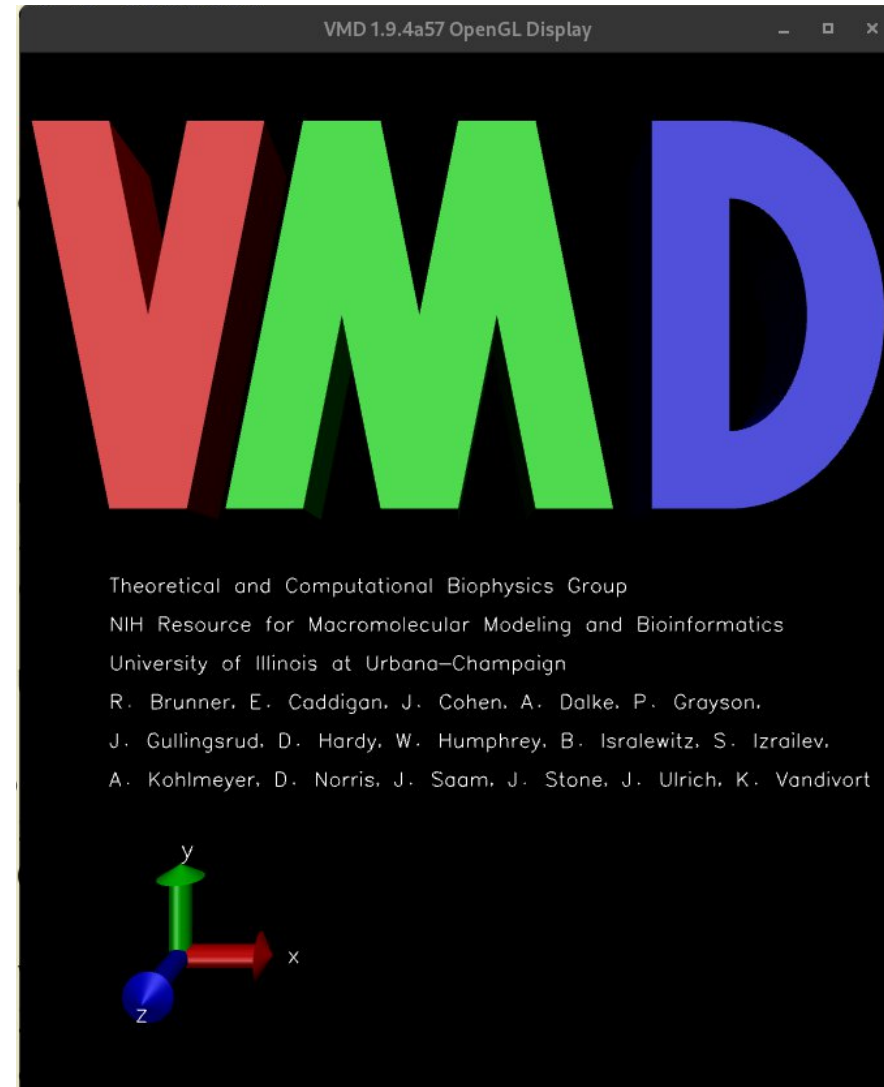
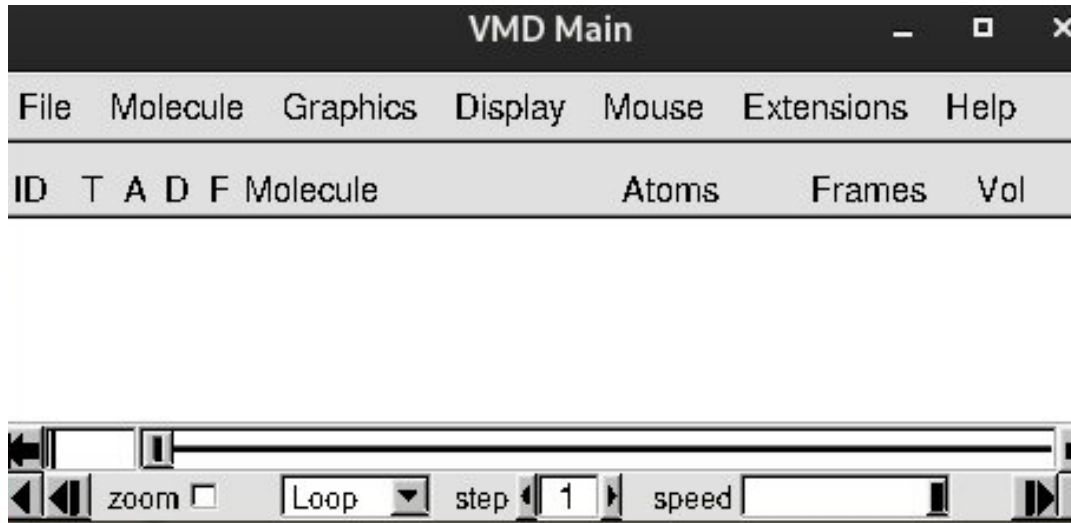
Launch

* The Remote Desktop Container session data for this session can be accessed under the [data root directory](#).

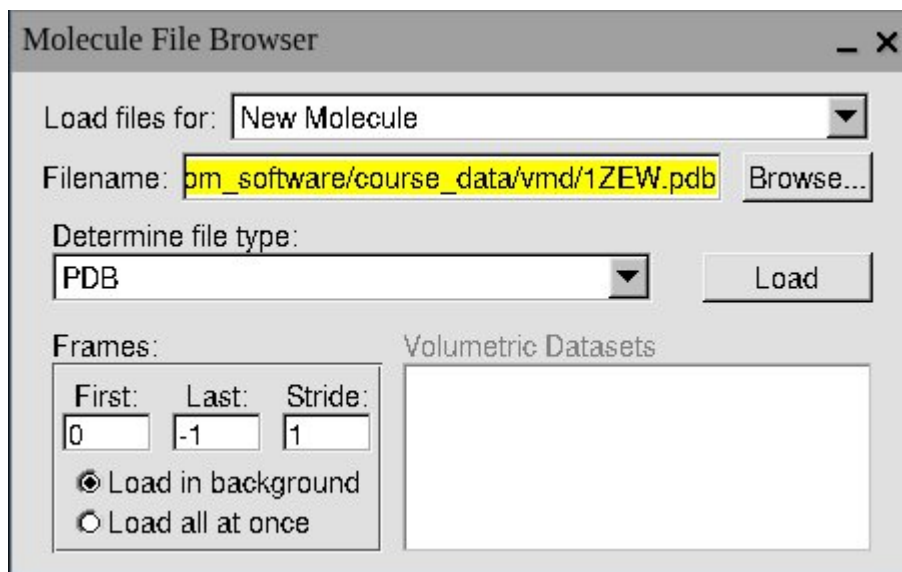
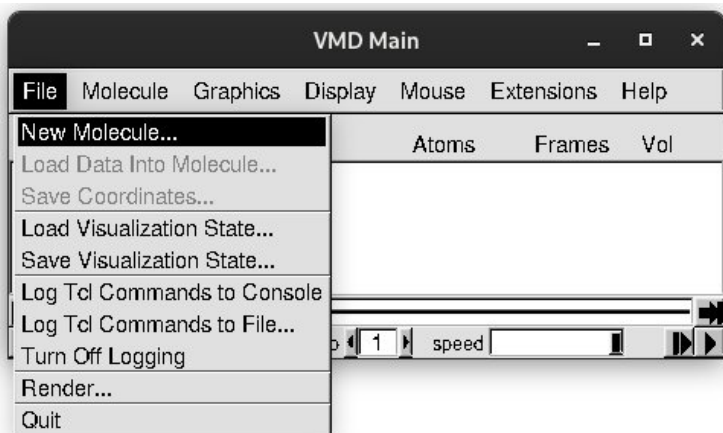


```
In addition to EESSI, other software  
e at /custom_software_rocky/spack/sha  
ux-rocky9-icelake  
  
Course data is available at /course_  
  
Backup your data to folder /backups/  
up only required data.  
  
[di97gef@node1 ~]$ module load vmd  
[di97gef@node1 ~]$ vmd
```

VMD in this course



Supported Formats



Automatically
 (Animated) XCrySDen Structure File
 ABINIT
 AMBER Coordinates
 AMBER Coordinates with Periodic Box
 AMBER Parm
 AMBER7 Parm
 AMBER7 Restart
 AVS Field
 Amira File Format (.am)
 Autodock Grid Map
 BRIX Density Map
 Basis Set
 Biomocca Volumetric Map
 CCP4, MRC Density Map
 CHARMM Coordinates
 CHARMM PBEQ Binary Potential Map
 CHARMM,NAMD,XPLOR DCD Trajectory
 CHARMM,NAMD,XPLOR PSF
 CPMD
 DESRES Molecular Structure
 DESRES Trajectory
 DL_POLY CONFIG
 DL_POLY_4 HISTORY
 DL_POLY_C HISTORY
 DSN6
 DX
 Delphi 'Big' Formatted Potential Map
 FS4 Density Map
 GAMESS
 GRASP
 GRASP,Delphi Binary Potential Map
 GRID,UHBD Binary Potential Map
 Gaussian Cube
 Gromacs GRO
 Gromacs TRJ Trajectory
 Gromacs TRR Trajectory
 Gromacs XTC Compressed Trajectory
 Gromacs g96
 HOOMD-blue XML File
 InsightII MDF
 InsightII car

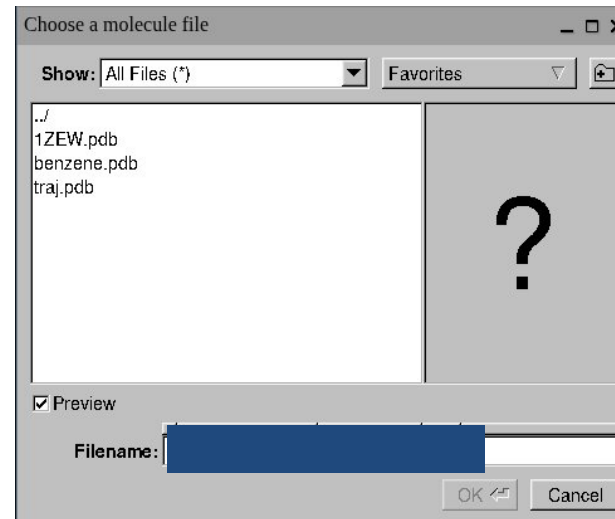
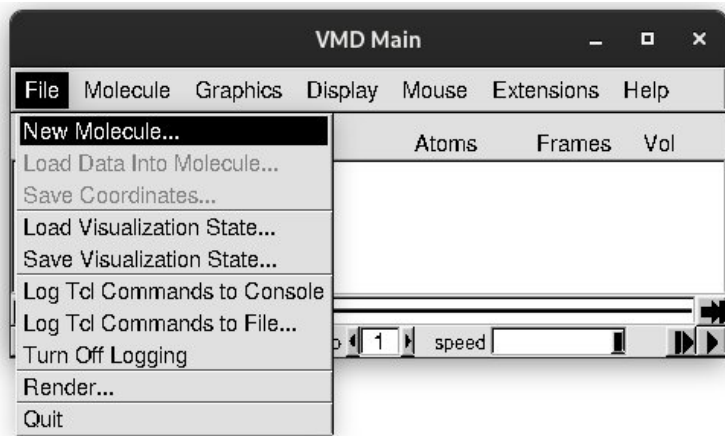
HOOMD-blue XML File
 InsightII MDF
 InsightII car
 Internal Paratool Format
 LAMMPS Trajectory
 Lattice Microbes (RDME)
 MDL mol2
 MSI Biograf Format
 MSMS Surface Mesh
 Maestro File
 Molden
 NAMD Binary Coordinates
 NetCDF (AMBER, MMTK)
 Object File Format (OFF)
 Orca

PDB

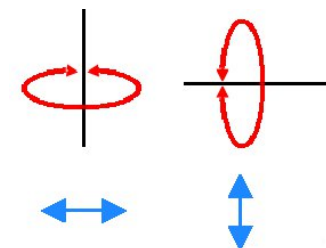
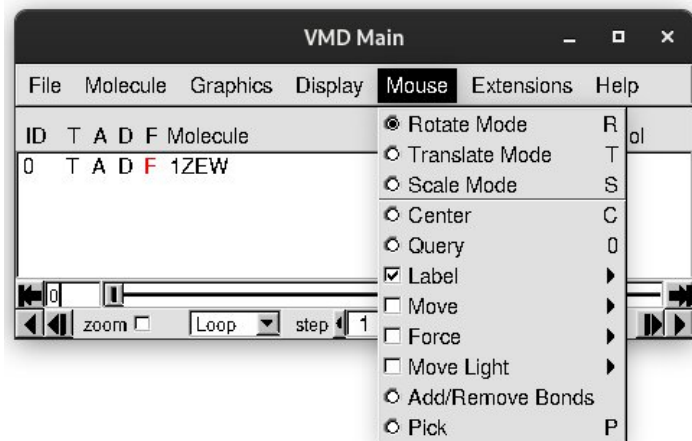
PLY
 PQR
 QCSchema
 Raster3d Scene File
 SPIDER Density Map
 STL Stereolithography Triangle Mesh
 Scripps Binpos
 Situs Density Map
 Tinker
 UHBD Grid
 VASP_CHGCAR
 VASP_OUTCAR
 VASP_PARCHG
 VASP_POSCAR
 VASP_XDATCAR
 VASP_XDATCAR5
 VASP_xml
 VTF coordinate format
 VTF structure format
 VTF trajectory format
 VTK grid reader
 Web PDB Download
 XPLOR Electron Density Map
 XYZ
 gOpenmol plt
 js
 mmCIF
 polygon mesh
 Convert from:

Interacting with molecule

BioPhyCourse-latest/Day2/01_VMD/data/1ZEW.pdb



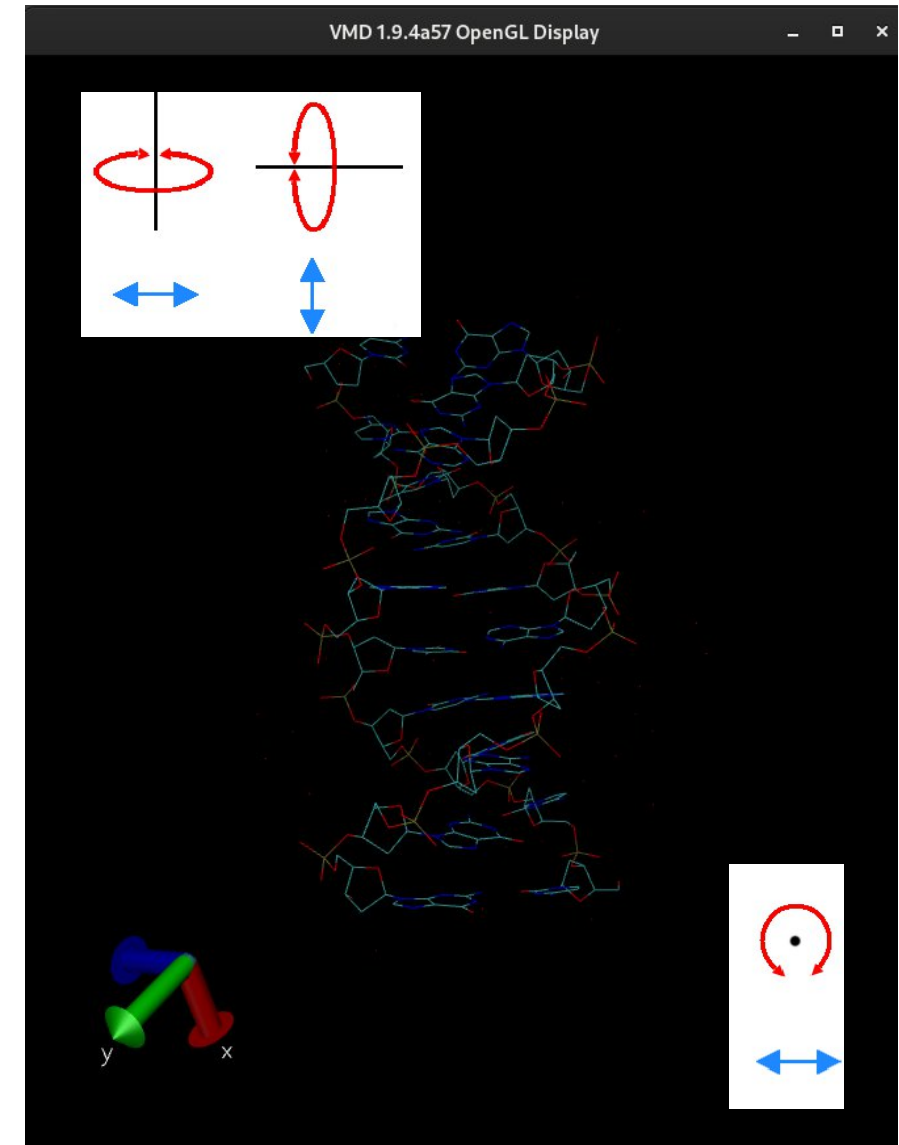
Rotate, translate and scale



Left Mouse

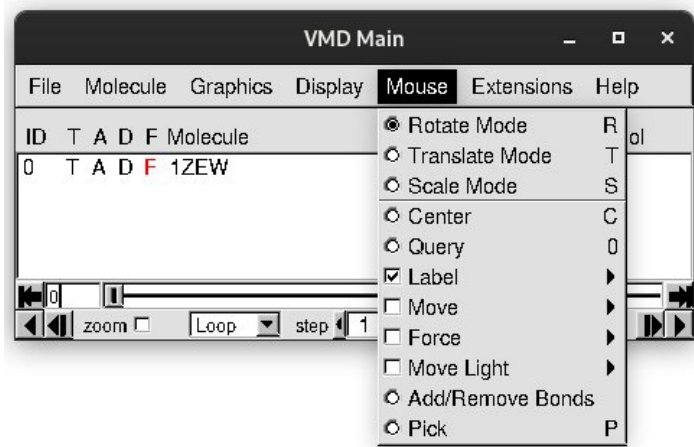


R M



Interacting with molecule

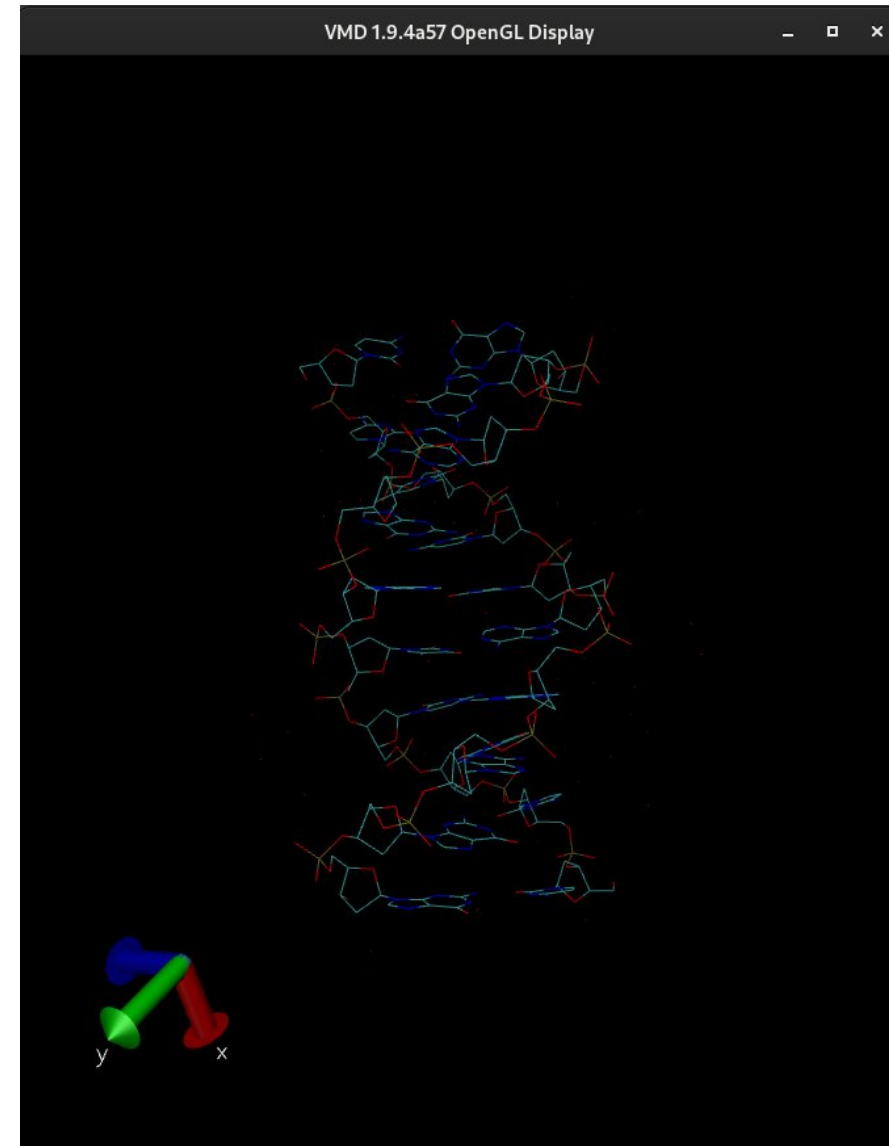
Translate and scale



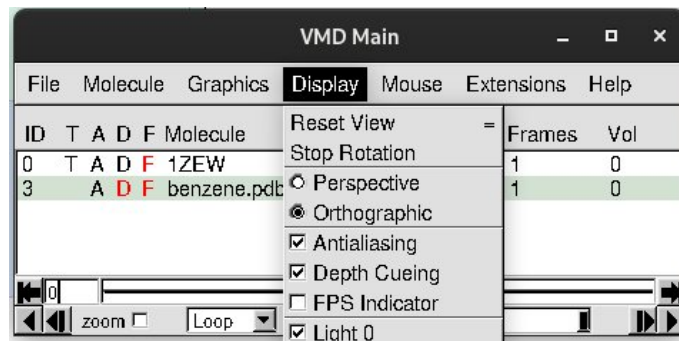
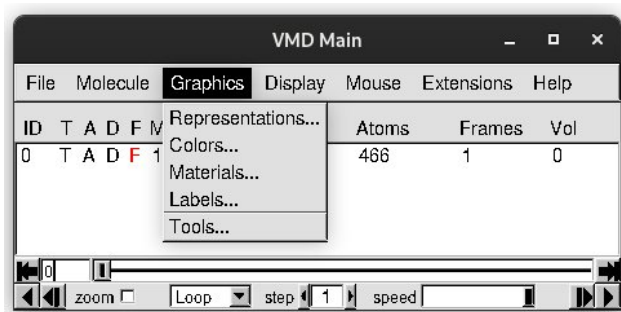
Press T to Translate -



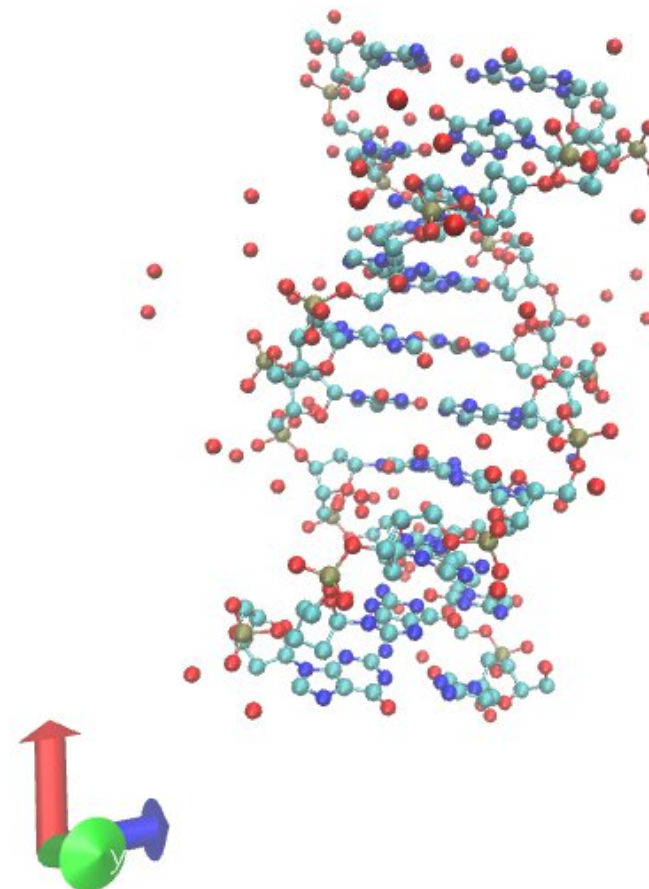
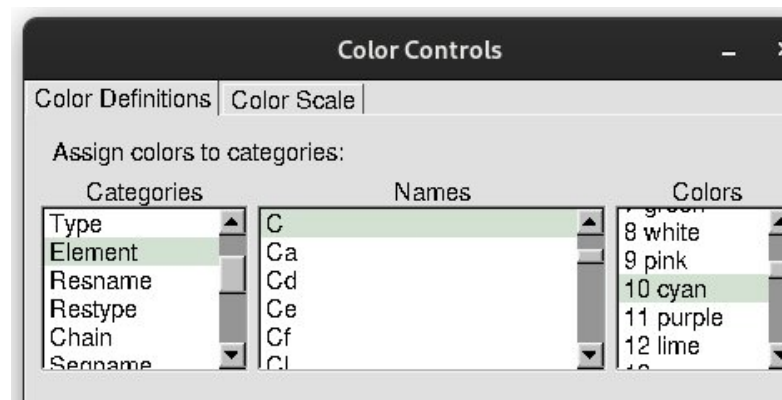
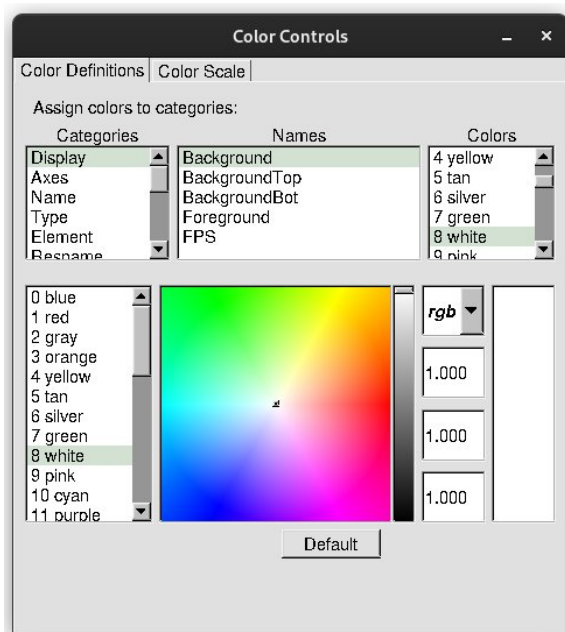
Press S to Scale -



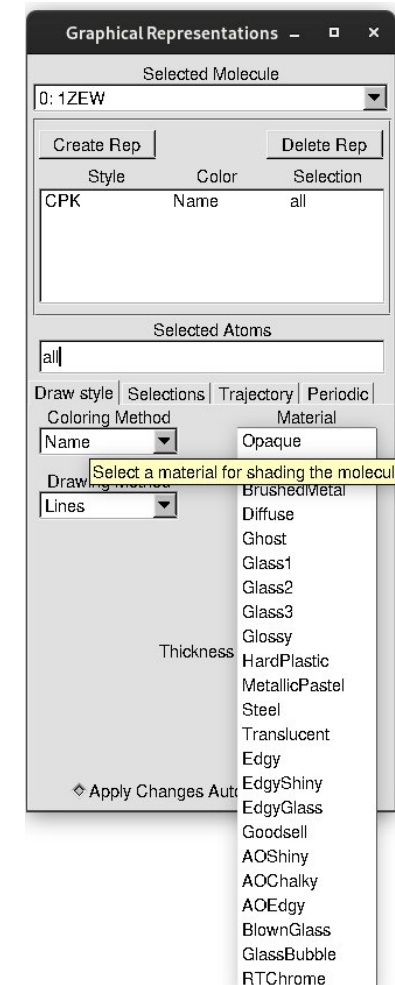
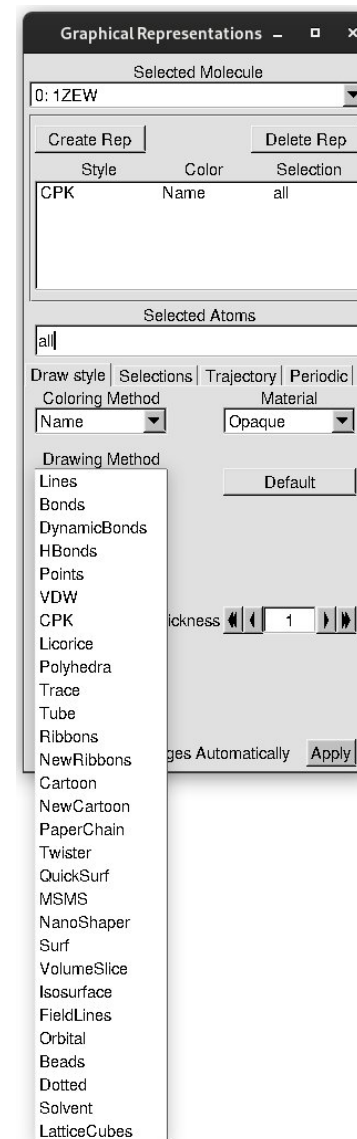
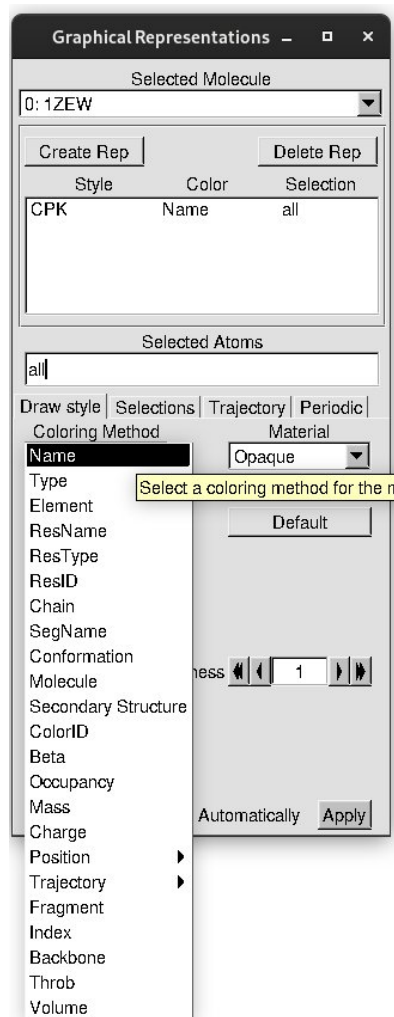
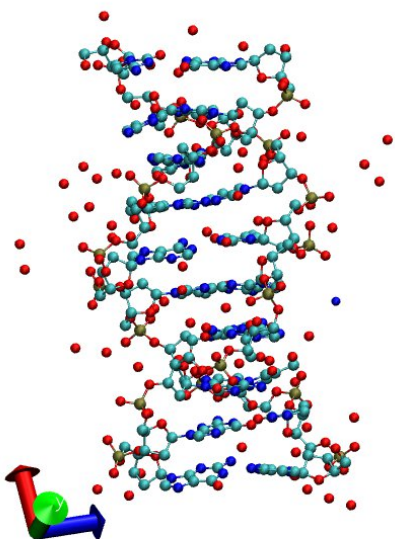
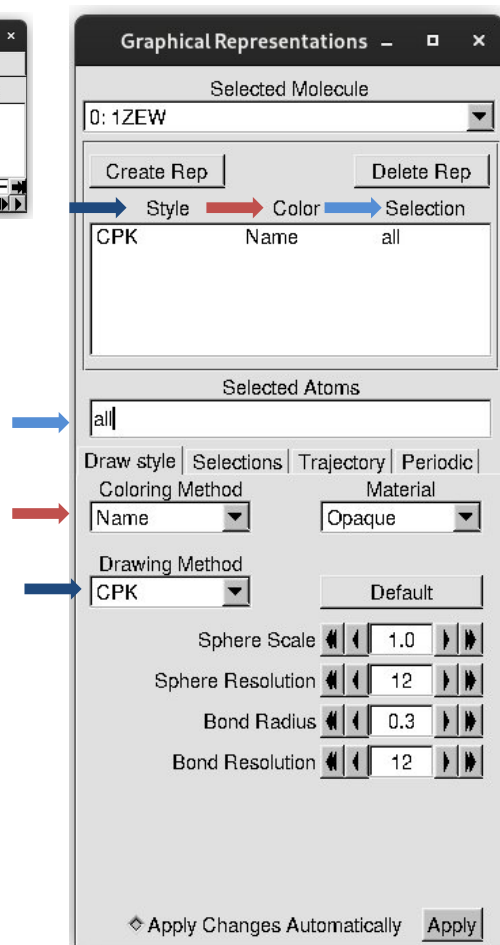
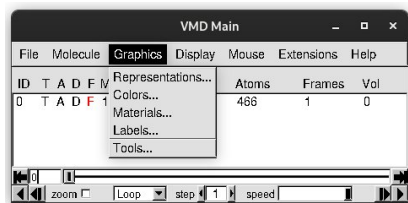
Color Settings



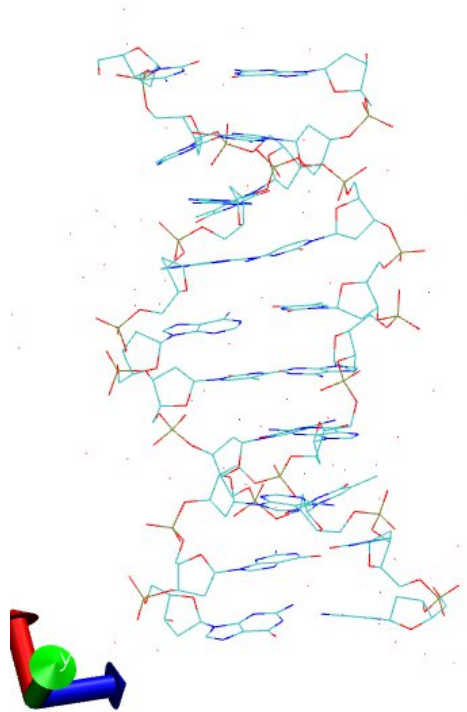
change colors of background, axes, elements.....



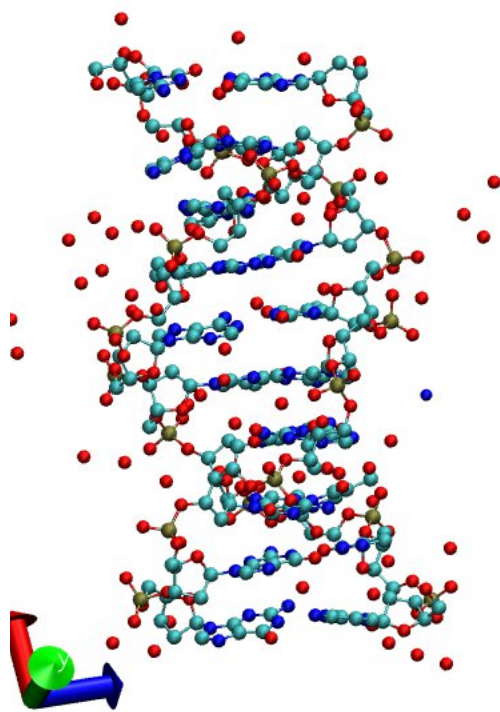
Graphical Representation



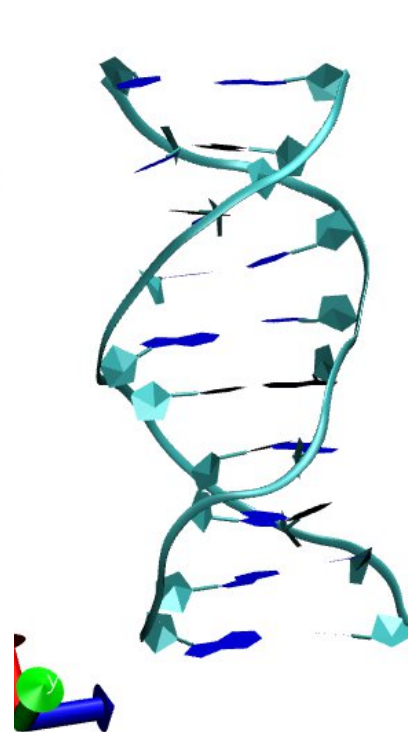
Graphical Representation: Drawing Method



Lines



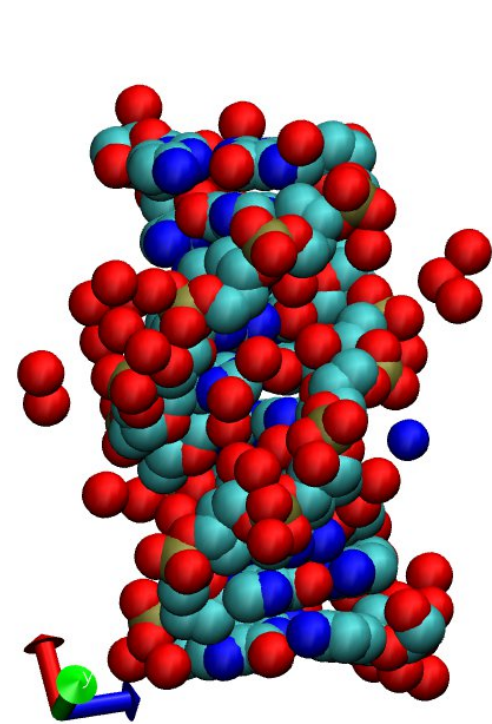
CPK



New Ribbons



New Cartoon



VDW

Graphical Representation: Selected Atoms

Graphical Representations

Selected Molecule

0: 1ZEW

Create Rep

Delete Rep

Style

Color

Selection

CPK

Name

all

Selected Atoms

all

Draw style

Selections

Trajectory

Periodic

Coloring Method

Material

Name

Opaque

Drawing Method

Default

Sphere Scale

1.0

Sphere Resolution

12

Bond Radius

0.3

Bond Resolution

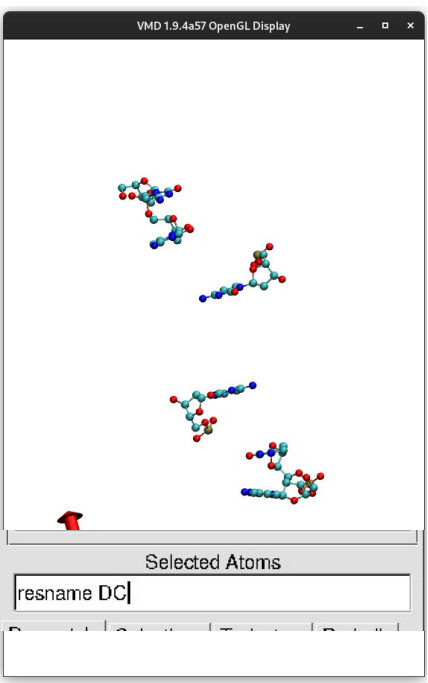
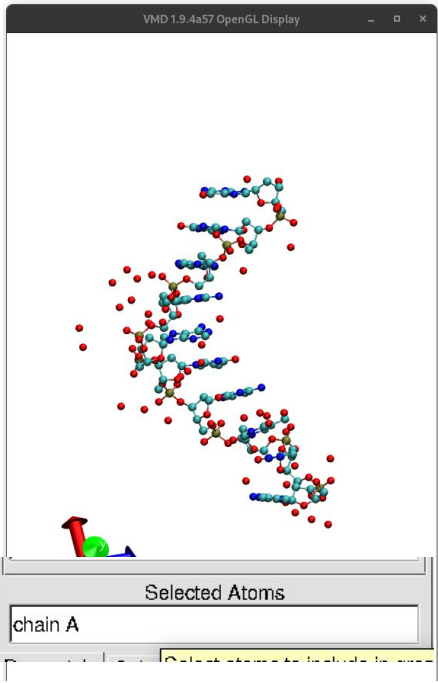
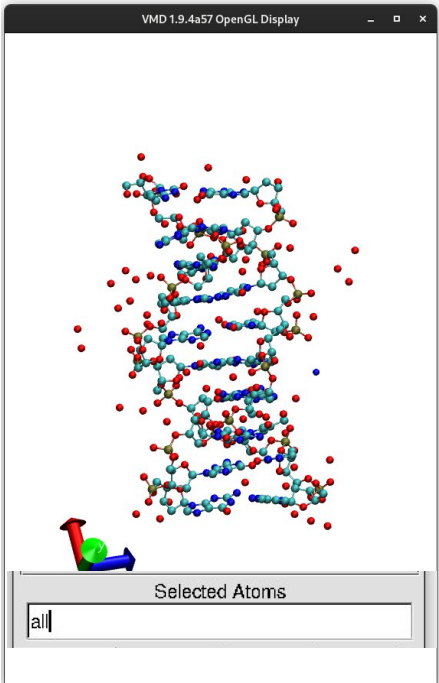
12

Apply Changes Automatically

Apply

	Serial	type	res	chain	coordinates							
ATOM	1	O5'	DC	A	1	-0.633	0.238	32.103	1.00	23.11		O
ATOM	2	C5'	DC	A	1	-1.560	-0.576	32.829	1.00	22.85		C
ATOM	3	C4'	DC	A	1	-1.619	-1.987	32.293	1.00	22.73		C
ATOM	4	O4'	DC	A	1	-2.109	-1.980	30.932	1.00			O
ATOM	5	C3'	DC	A	1	-0.282	-2.719	32.253	1.00			C

PDB File



- index 1- 100
 - resname DC
 - chain A
- x > 10
 - y < 20
 - x > 10 && y < 20
- element C
 - type O5'
 - water

Columns	Data
1-4	"ATOM"
7-11#	Atom serial number
13-16	Atom name
17	Alternate location indicator
18-20\$	Residue name
22	Chain identifier
23-26	Residue sequence number
27	Code for insertions of residues
31-38	X orthogonal Å coordinate
39-46	Y orthogonal Å coordinate
47-54	Z orthogonal Å coordinate
55-60	Occupancy
61-66	Temperature factor
73-76	Segment identifier¶
77-78	Element symbol
79-80	Charge

Graphical Representation: Create Rep

Three screenshots of the VMD 'Graphical Representations' window showing the process of creating a new representation for the selected molecule (0: 1ZEW).

Screenshot 1: The 'Create Rep' button is highlighted. The table below shows the current state of the representations:

Style	Color	Selection
CPK	Name	all

Screenshot 2: The 'Create Rep' button is highlighted. The table below shows the current state of the representations:

Style	Color	Selection
CPK	Name	all
CPK	Name	all

Screenshot 3: The 'Create Rep' button is highlighted. The table below shows the current state of the representations:

Style	Color	Selection
CPK	Name	all
NewCartoon	Name	all

The 'NewCartoon' representation is selected. The 'Drawing Method' is set to 'NewCartoon' and the 'Spline Style' is set to 'Catmull-Rom'.

The resulting 3D visualization shows a DNA double helix structure rendered using the 'NewCartoon' drawing method, with atoms represented by spheres and bonds by thick, curved lines. A coordinate system is visible at the bottom left.

Graphical Representation: Create Rep

Diagram illustrating the process of creating a graphical representation (Rep) in VMD, showing three sequential screenshots of the 'Graphical Representations' window and the resulting 3D molecular model.

Step 1: Initial State

- Selected Molecule: 0: 1ZEW
- Buttons: Create Rep, Delete Rep
- Table:

Style	Color	Selection
CPK	Name	all

- Selected Atoms: all
- Draw style: Name
- Coloring Method: Opaque
- Drawing Method: CPK
- Parameters: Sphere Scale (1.0), Sphere Resolution (12), Bond Radius (0.3), Bond Resolution (12)
- Buttons: Apply Changes Automatically, Apply

Step 2: Adding a New Representation

- Selected Molecule: 0: 1ZEW
- Buttons: Create Rep, Delete Rep
- Table:

Style	Color	Selection
CPK	Name	all
CPK	Name	all

- Selected Atoms: all
- Draw style: Name
- Coloring Method: Opaque
- Drawing Method: CPK
- Parameters: Sphere Scale (1.0), Sphere Resolution (12), Bond Radius (0.3), Bond Resolution (12)
- Buttons: Apply Changes Automatically, Apply

Step 3: Final State (Selected Atoms: chain A)

- Selected Molecule: 0: 1ZEW
- Buttons: Create Rep, Delete Rep
- Table:

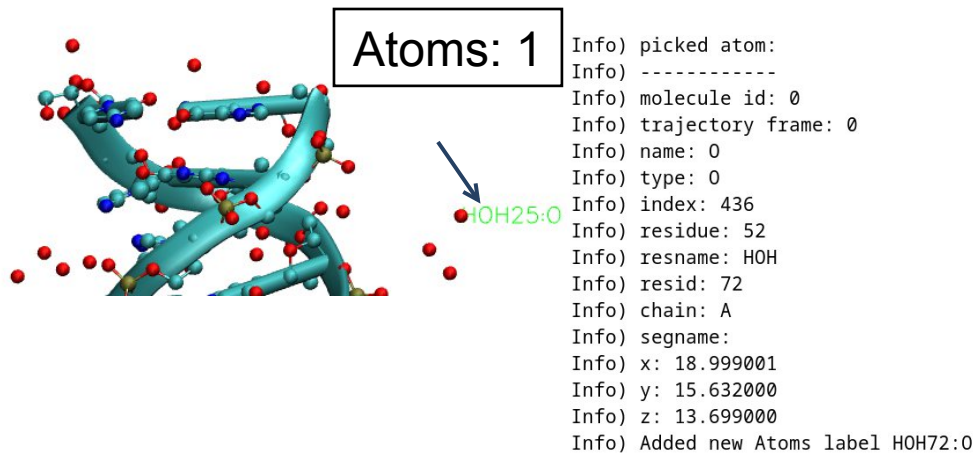
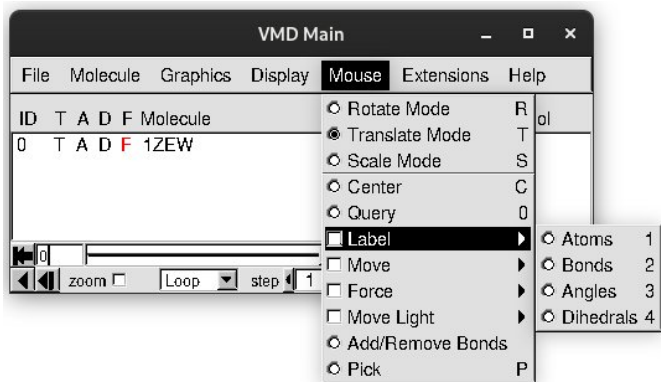
Style	Color	Selection
CPK	Name	chain A
NewCartoon	ColorID 12	chain B

- Selected Atoms: chain A
- Draw style: Name
- Coloring Method: AOShiny
- Drawing Method: CPK
- Parameters: Sphere Scale (1.0), Sphere Resolution (12), Bond Radius (0.3), Bond Resolution (12)
- Buttons: Apply Changes Automatically, Apply

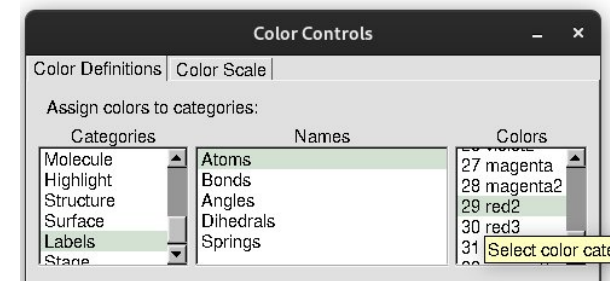
Resulting 3D Model:

The final 3D molecular model shows the protein structure (chain A) rendered using the 'NewCartoon' style with 'ColorID 12' (green), and the surrounding water molecules (chain B) rendered using the 'CPK' style with 'Name' (red, blue, and white spheres).

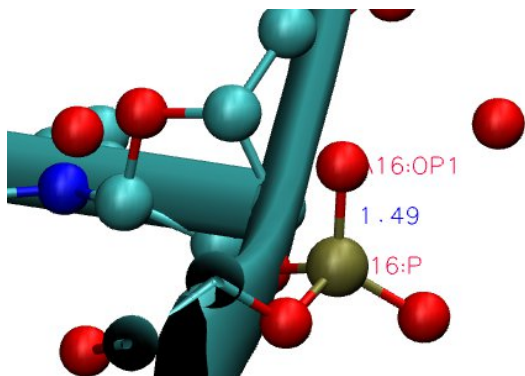
Label: Atoms, Bonds, Angles and Dihedrals



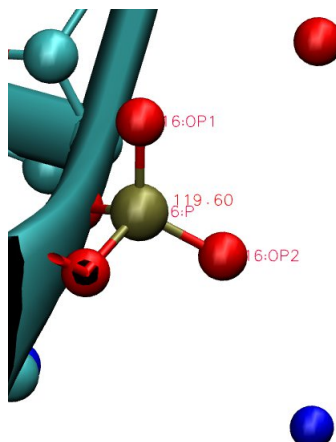
change colors



Bonds: 2

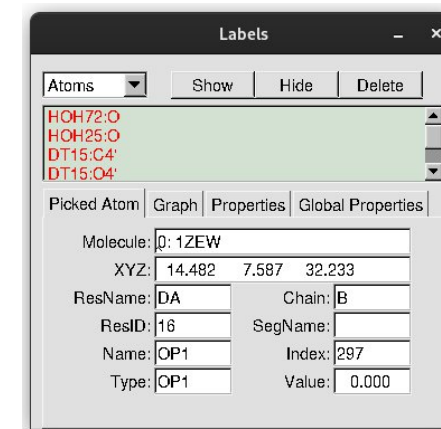


Angles: 3



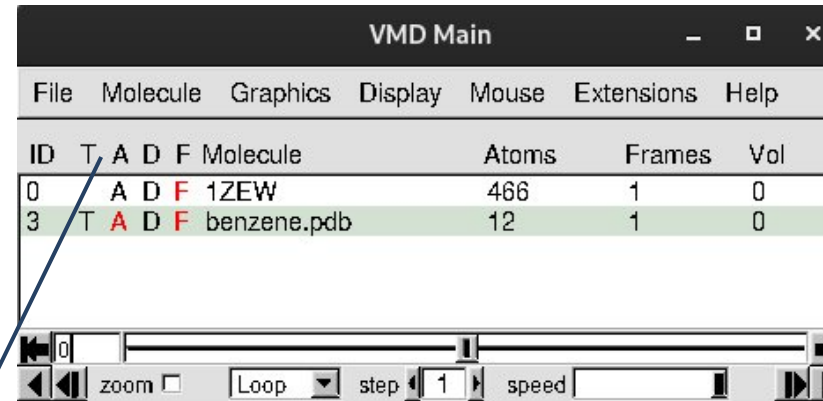
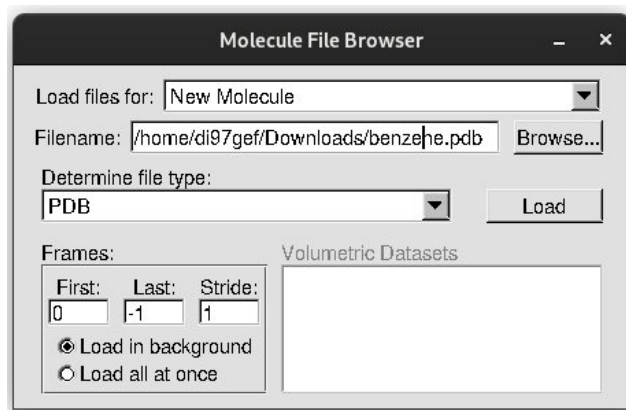
Delete Labels

Graphics > Labels

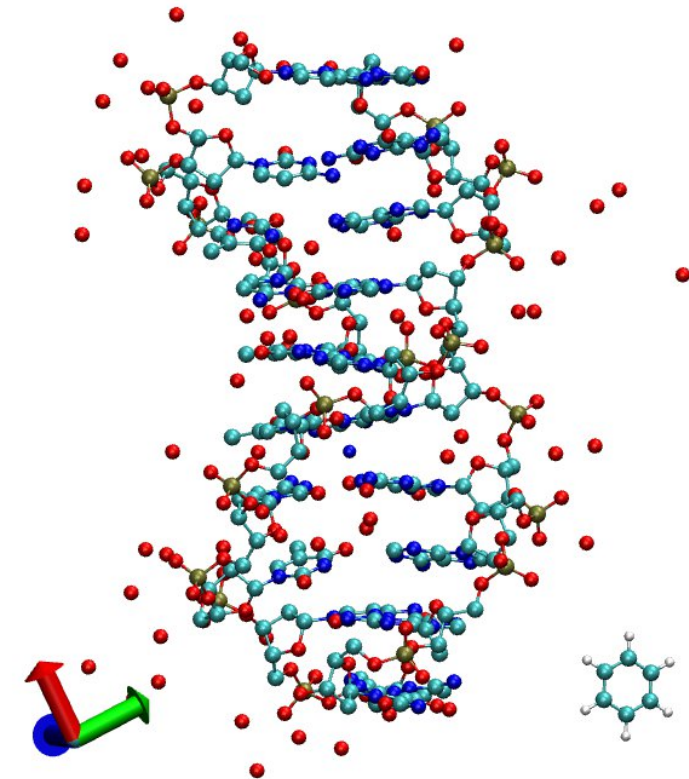
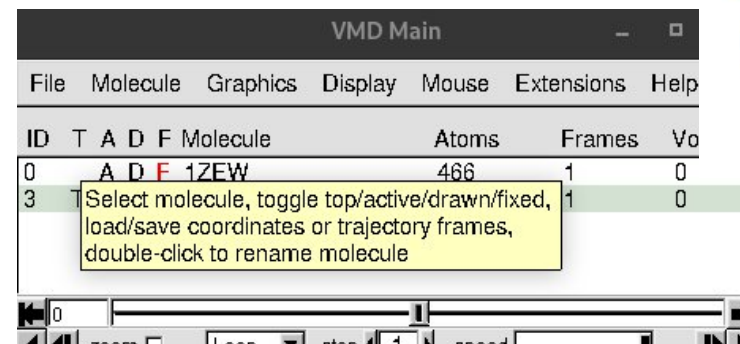


VMD Main: Multiple Molecules

Load new molecule



ID - Internal Molecule ID
 T - Top; All actions apply to top molecule
 A - Active; Trajectory update status
 D - Drawn
 F - Fixed



Rendering: Publication Quality Images

Change Material Type

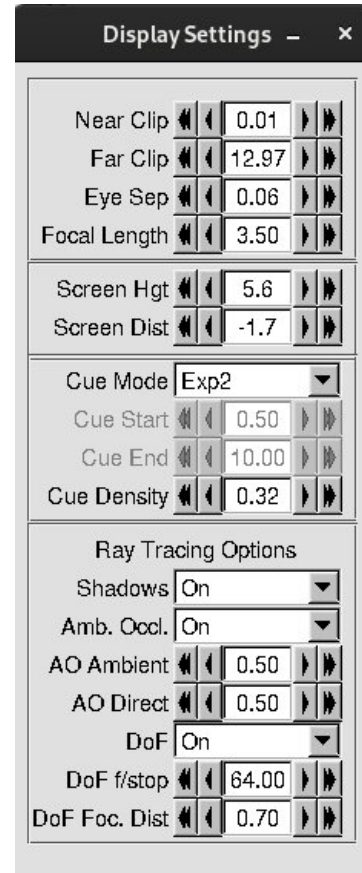
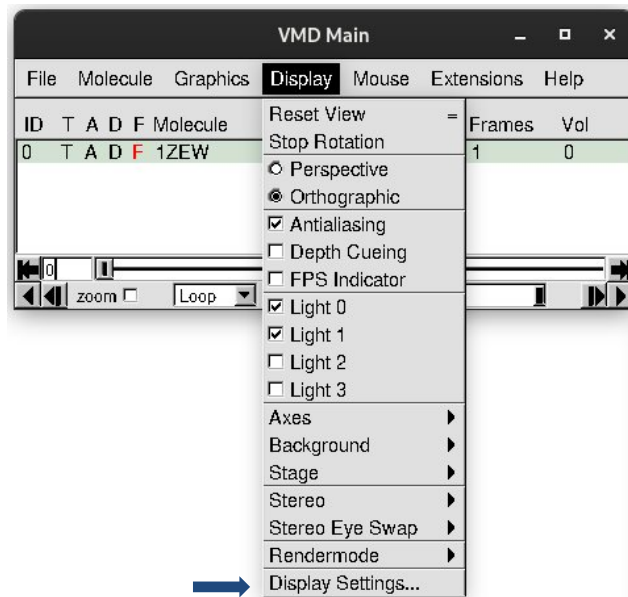
The first screenshot shows the 'Graphical Representations' window for molecule 0: 1ZEW. The 'Selected Atoms' field contains 'chain B'. The 'Coloring Method' is set to 'Name' and the 'Material' is 'AOShiny'. The 'Drawing Method' is 'CPK'. The second screenshot shows the same window with the 'Material' dropdown menu open, displaying a list of material options including Opaque, Transparent, BrushedMetal, Diffuse, Ghost, Glass1, Glass2, Glass3, Glossy, HardPlastic, MetallicPastel, Steel, Translucent, Edgy, EdgyShiny, EdgyGlass, Goodsell, AOCalky, AOEgdy, BlownGlass, GlassBubble, and RTChrome. The 'AOShiny' option is highlighted.

Change Resolution

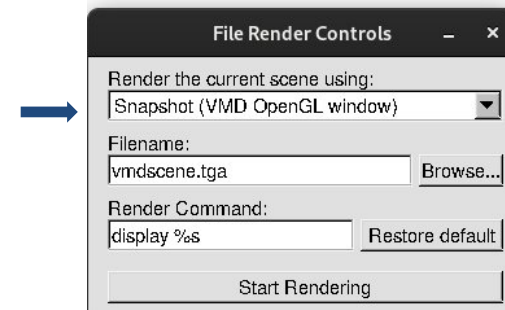
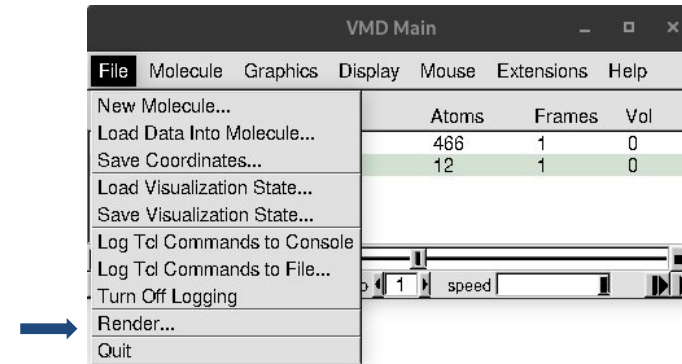
The third screenshot shows the 'Graphical Representations' window for molecule 0: 1ZEW. The 'Selected Atoms' field contains 'chain B'. The 'Coloring Method' is set to 'Name' and the 'Material' is 'AOCalky'. The 'Drawing Method' is 'NewCartoon'. The 'Sphere Resolution' is set to 12. The fourth screenshot shows the same window with the 'Sphere Resolution' slider moved to 200. The 'Bond Resolution' is also set to 200.

Rendering: Publication Quality Images

Go to Display Settings

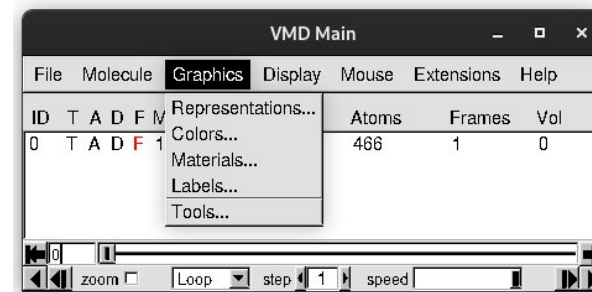


Go to render settings



- ART (VORT ray tracer)
- NVIDIA Gelato 2.1
- PostScript (vector graphics)
- Raster3D 2.7d
- Radiance 4.0
- Rayshade 4.0
- PIXAR RenderMan
- Snapshot (VMD OpenGL window)
- STL (triangle mesh only)
- Tachyon
- Tachyon (internal, in-memory rendering)**
- TachyonL-OSPRay (internal, in-memory, SSE+AVX-accelerated)
- TachyonL-OSPRay (interactive, SSE+AVX-accelerated)
- POV-Ray 3.6
- VRML 1.0 (VRML94)
- VRML 2.0 (VRML97)
- Wavefront (OBJ and MTL)
- X3D (XML) full specification
- X3D (XML) limited subset for X3DOM v1.1

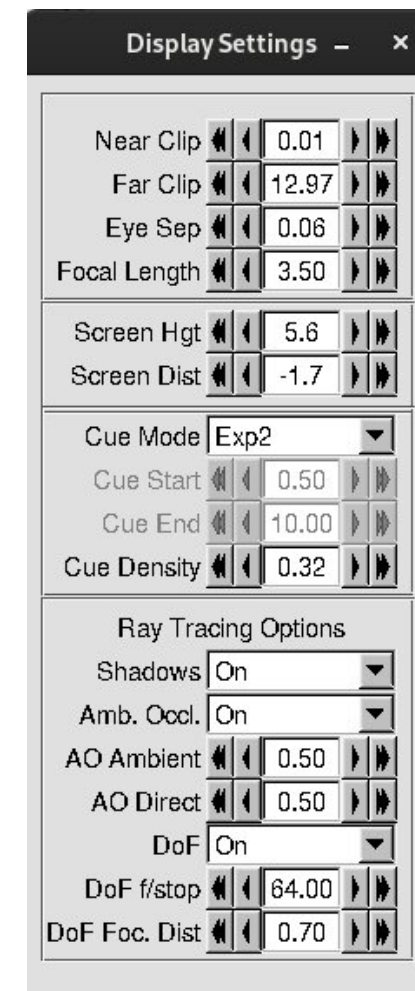
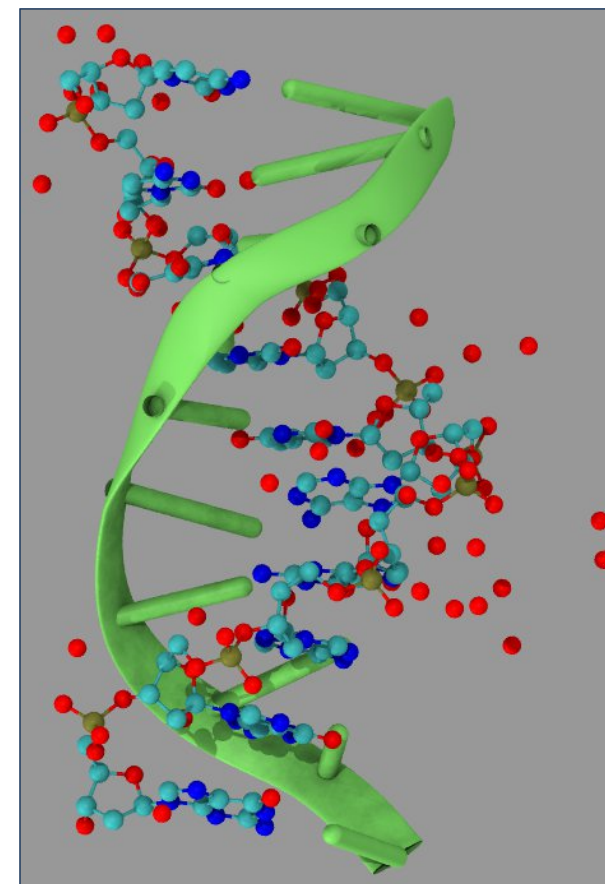
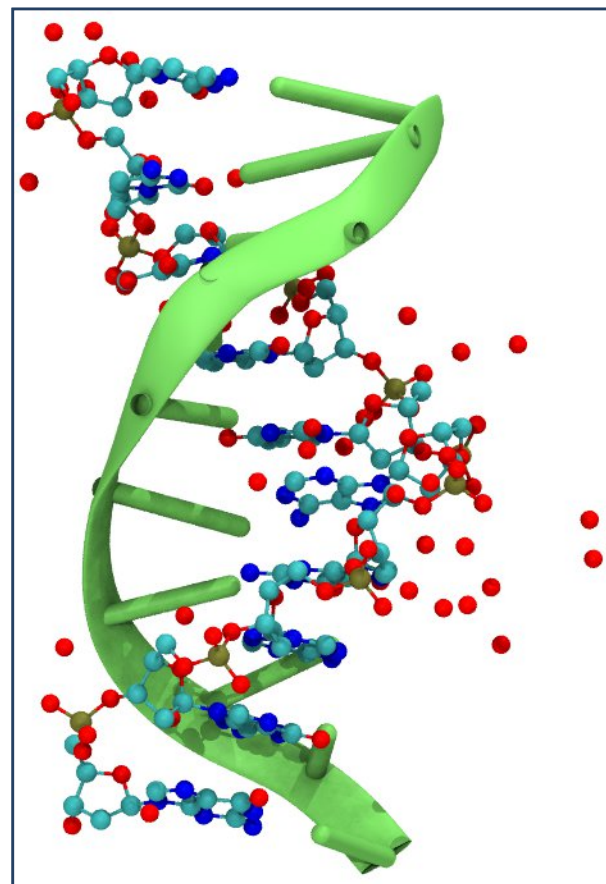
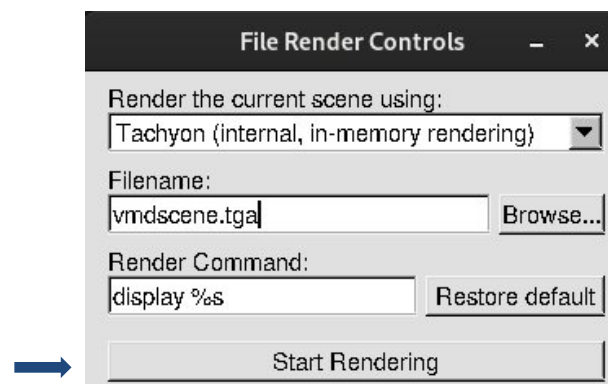
Rendering: Publication Quality Images



color > background > grey

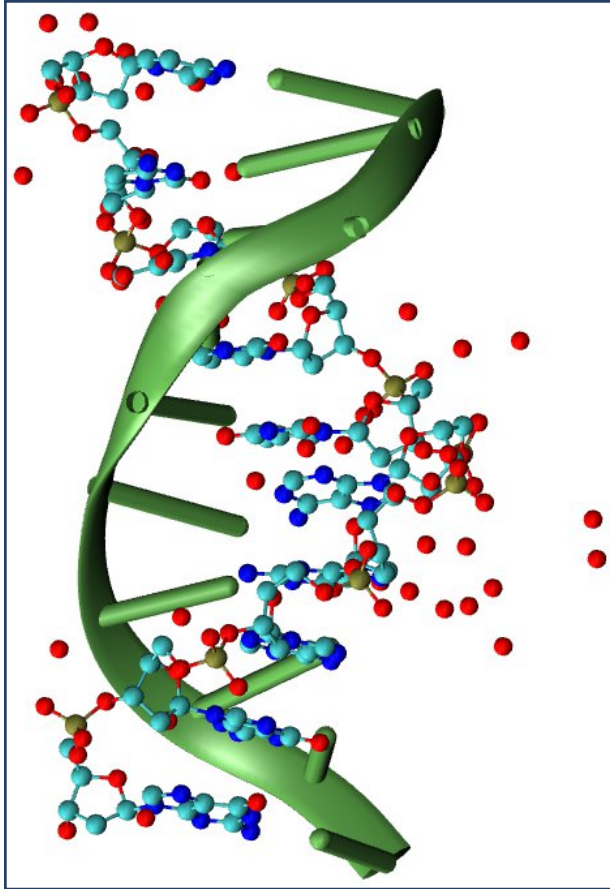
Fine tune Settings

Render Image

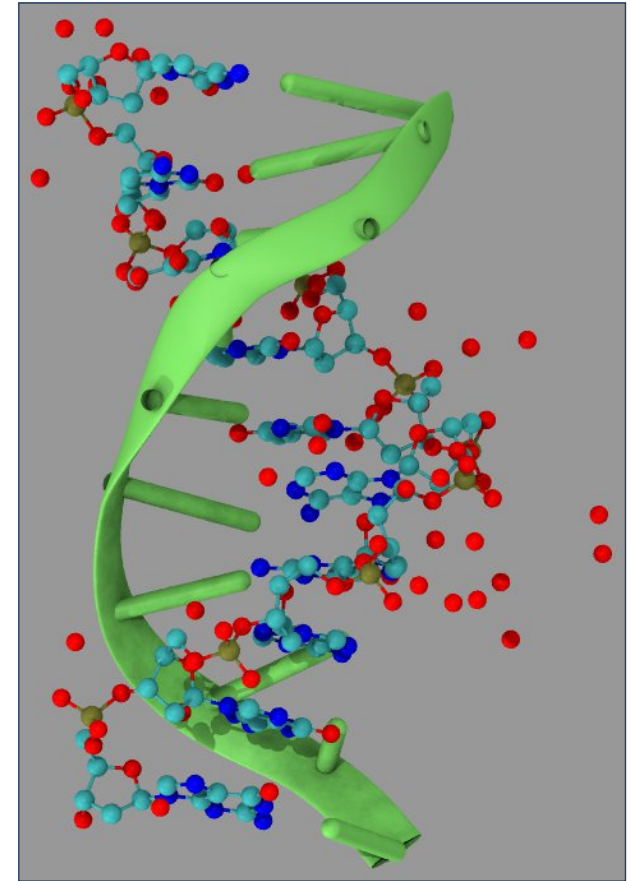
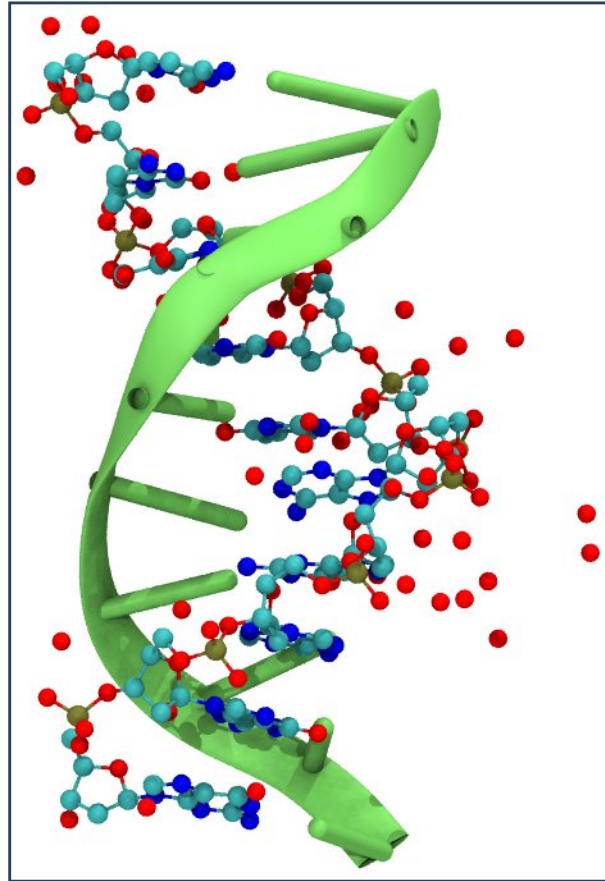


Rendering: Publication Quality Images

Using VMD Snapshot

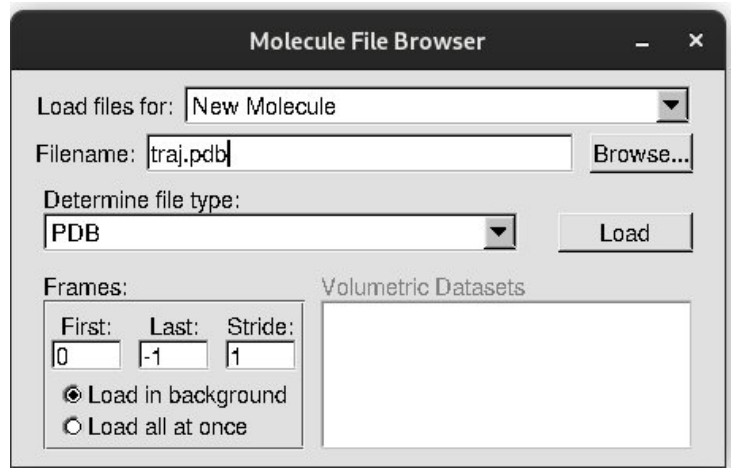


Using Tachyon Internal

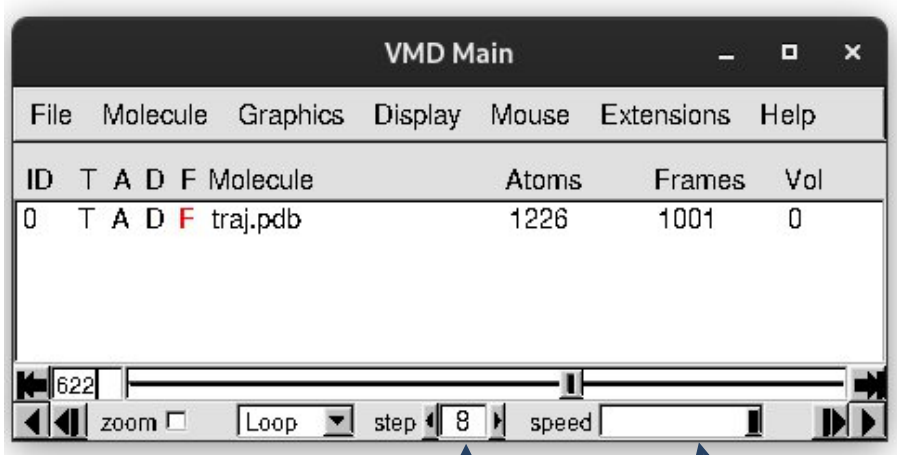


VMD: Loading Trajectories

Load trajectory



Load custom Frames

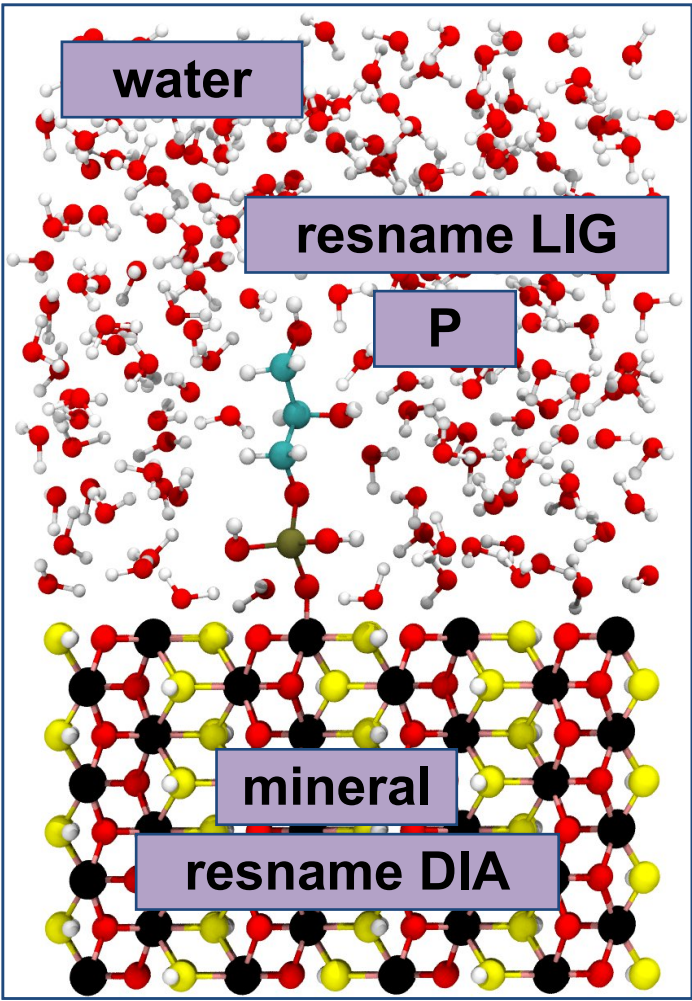


Loop/rock

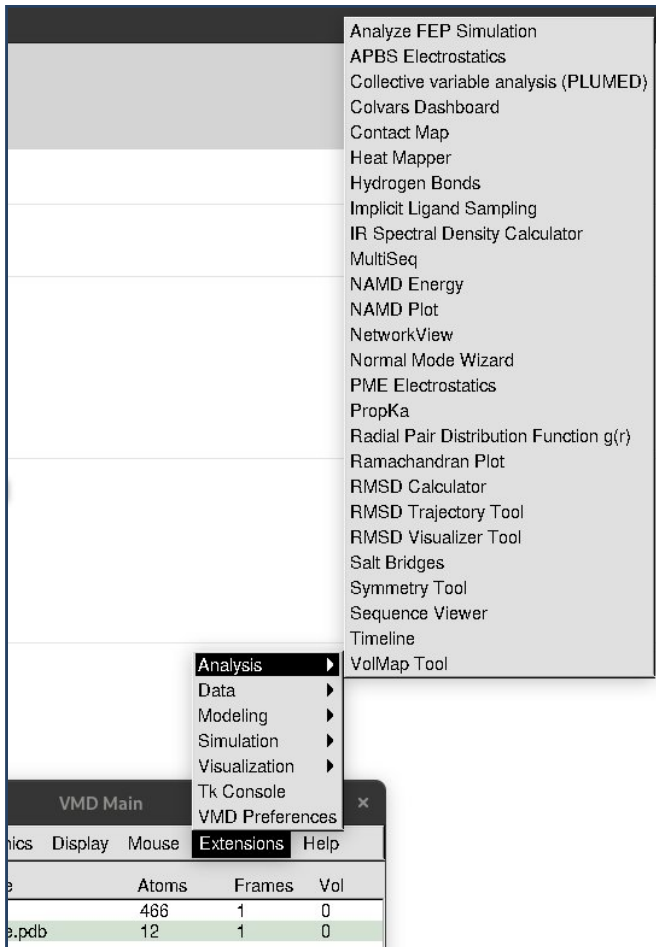
show every 8th step

Speed of frames

BioPhyCourse-latest/Day2/01_VMD/data/traj.pdb



VMD: Trajectory Analysis



Hydrogen Bond Analysis

Radial Pair Distribution Function

RMSD

VMD: Trajectory Analysis

Hydrogen Bond Analysis

Hydrogen Bonds

Input options

Molecule: 0: traj.pdb

Selection 1 (Required): resname LIG

Selection 2 (Optional): water

NOTE: if sel1 and sel2 overlap, hbonds output is unreliable!

Frames: all (now, all, b:e, or b:s:e)

☒ Update selections every frame? ☐ Only polar atoms (N, O, S, F)?

Selection 1 is the: ☐ Donor ☐ Acceptor ☒ Both

Donor-Acceptor distance (Å): 3.0

Angle cutoff (degrees): 40

Calculate detailed info for: ☒ None ☐ All hbonds ☐ Residue pairs ☐ Unique hbond

Output options

☒ Plot the data with MultiPlot?

Output directory: /home/di97get/Documents/lrz/work/biophy_cc Choose

Log file?

☐ Write output to files?

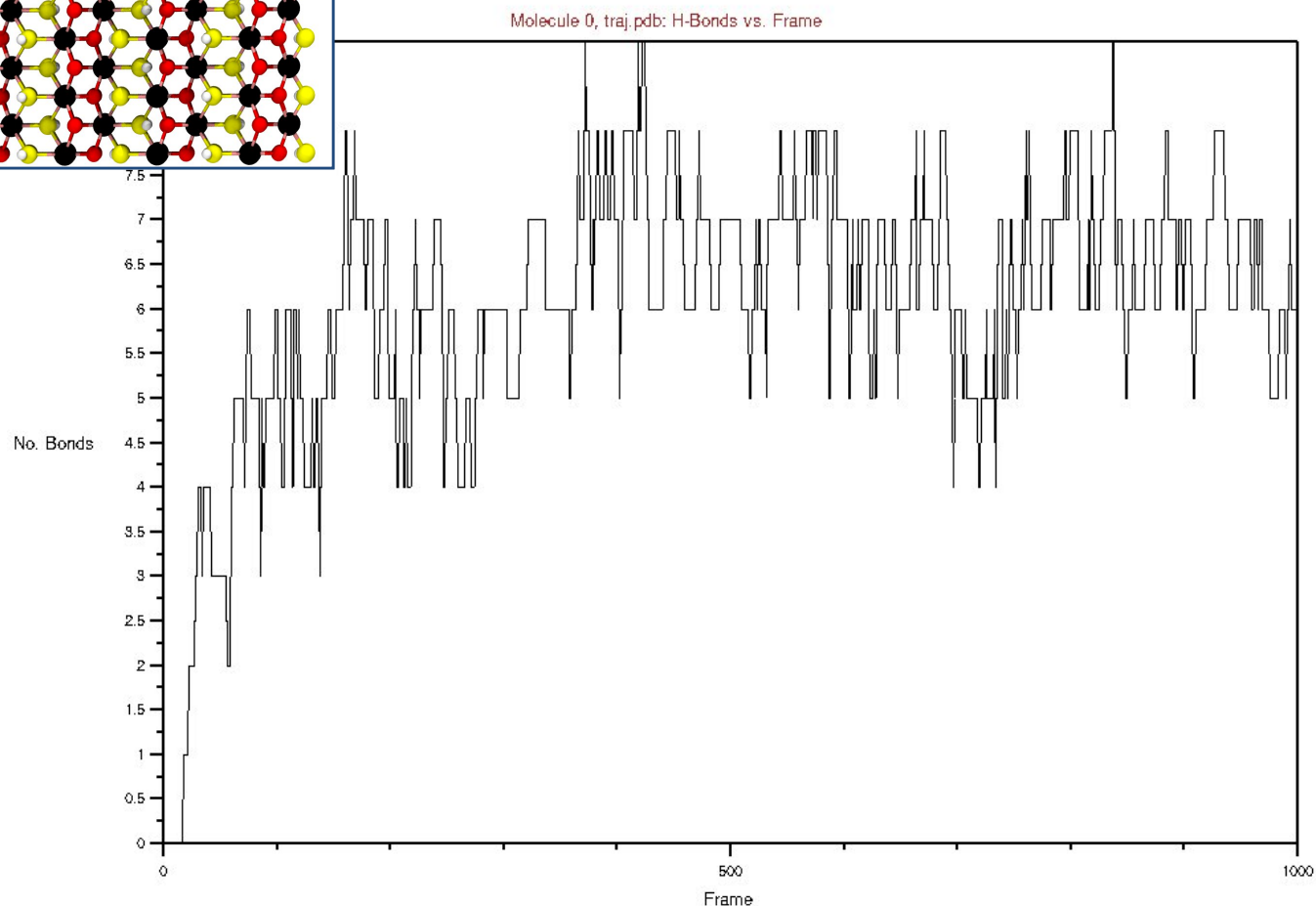
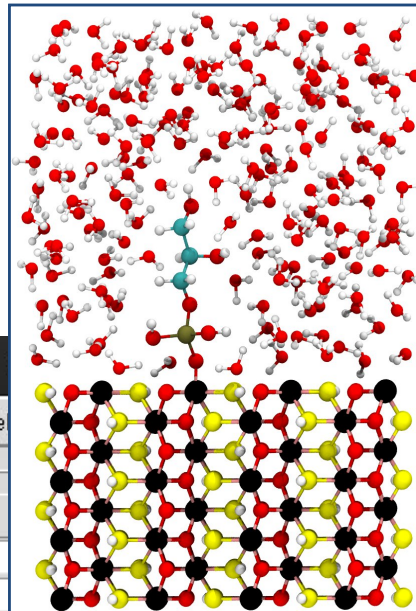
Frame/bond data? hbonds.dat

Detailed hbond data? hbonds-details.dat

Status

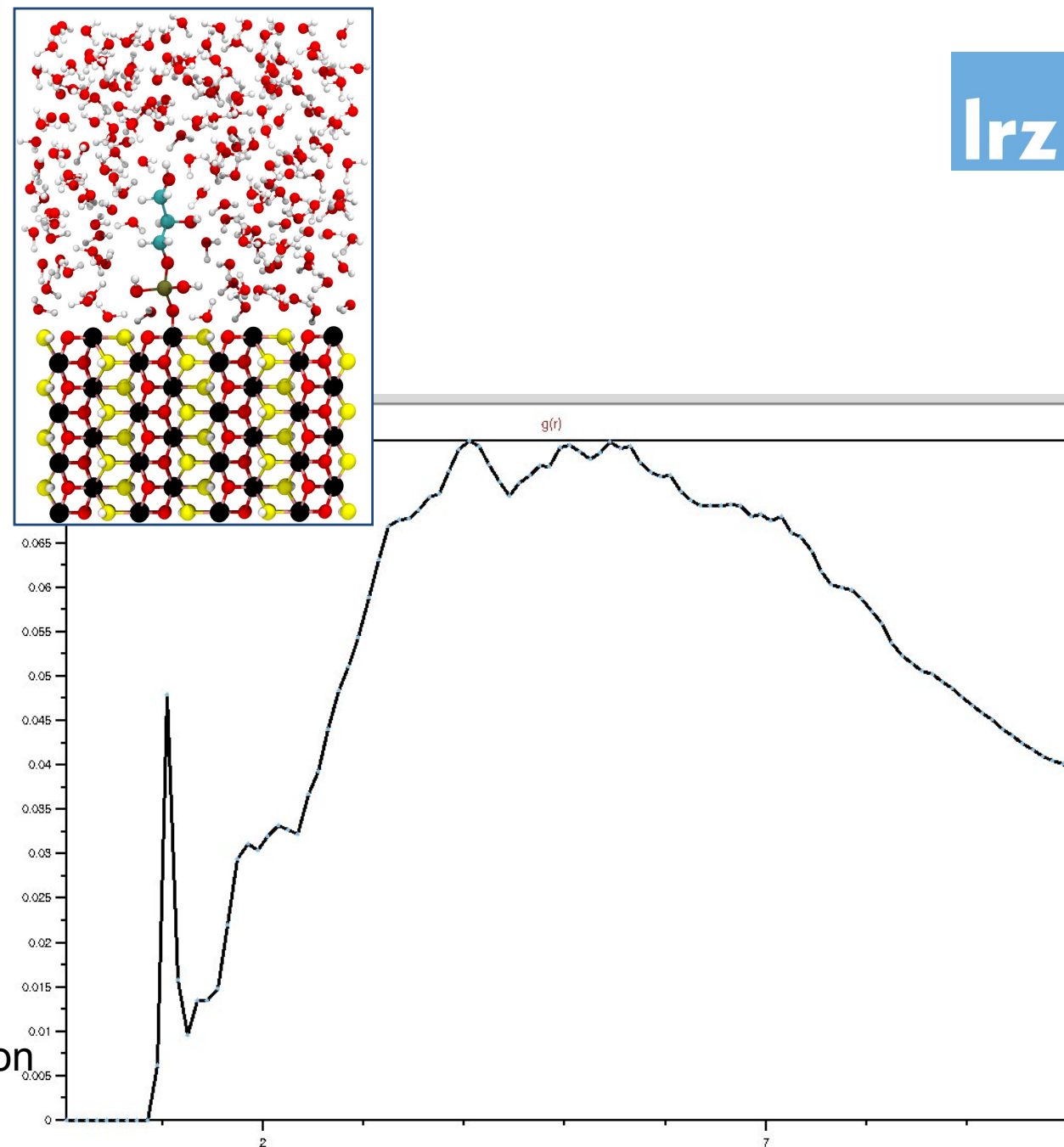
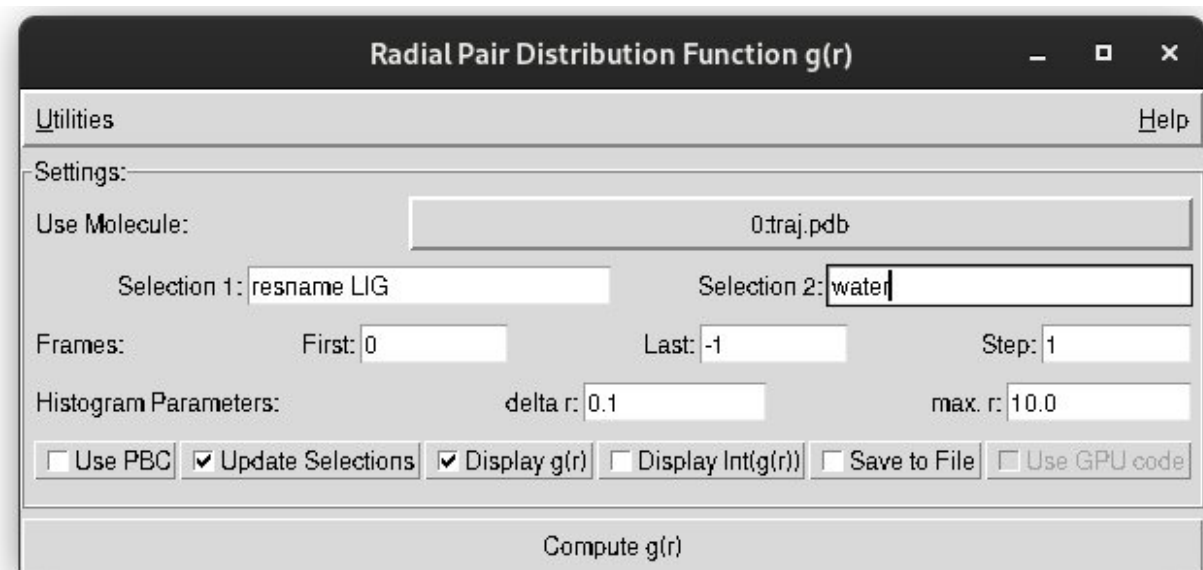
Ready.

Find hydrogen bonds!



VMD: Trajectory Analysis

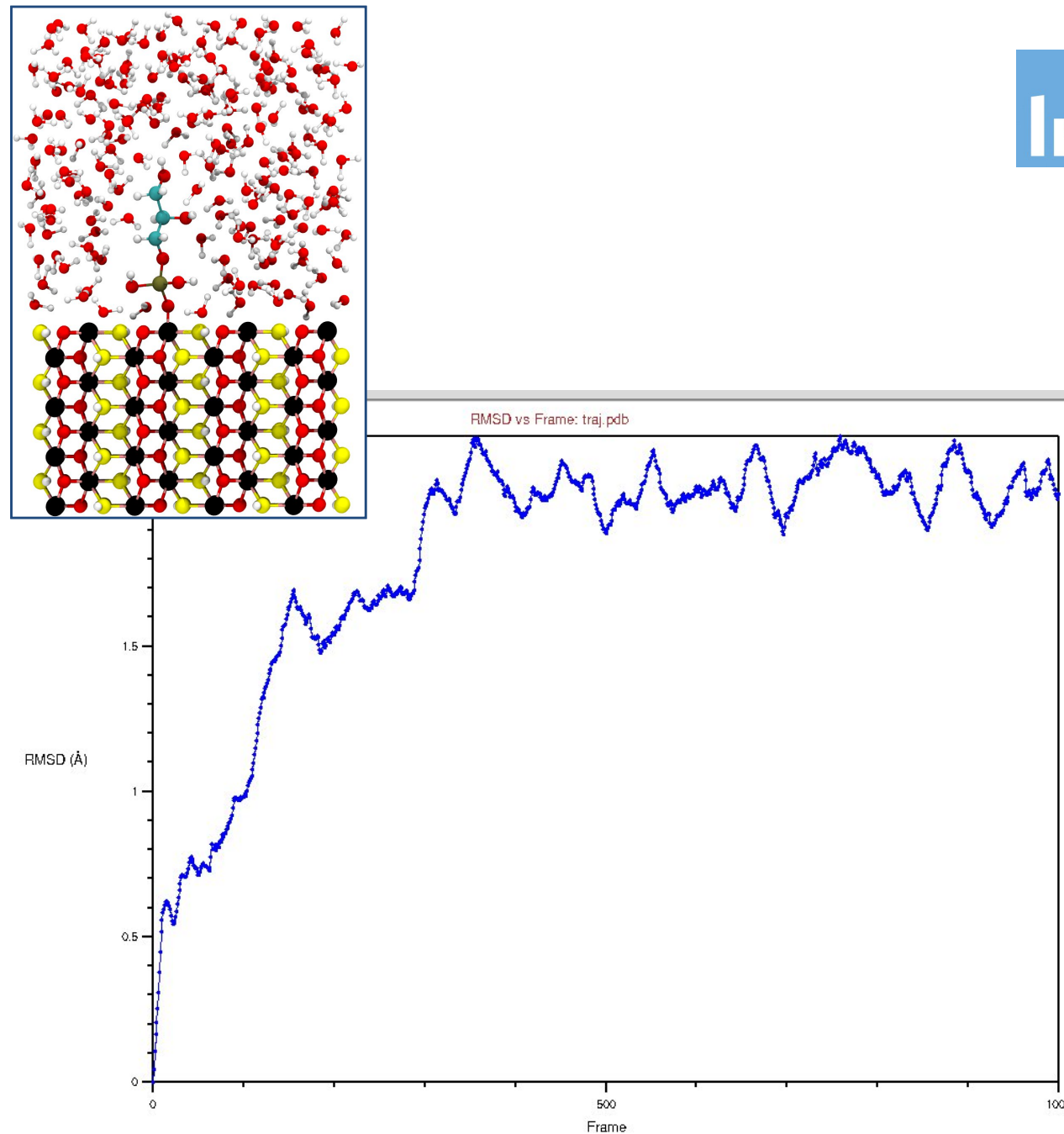
Radial Pair Distribution Function



Extensions > Analysis > Radial Pair Distribution Function

VMD: Trajectory Analysis

RMSD Visualizer Tool

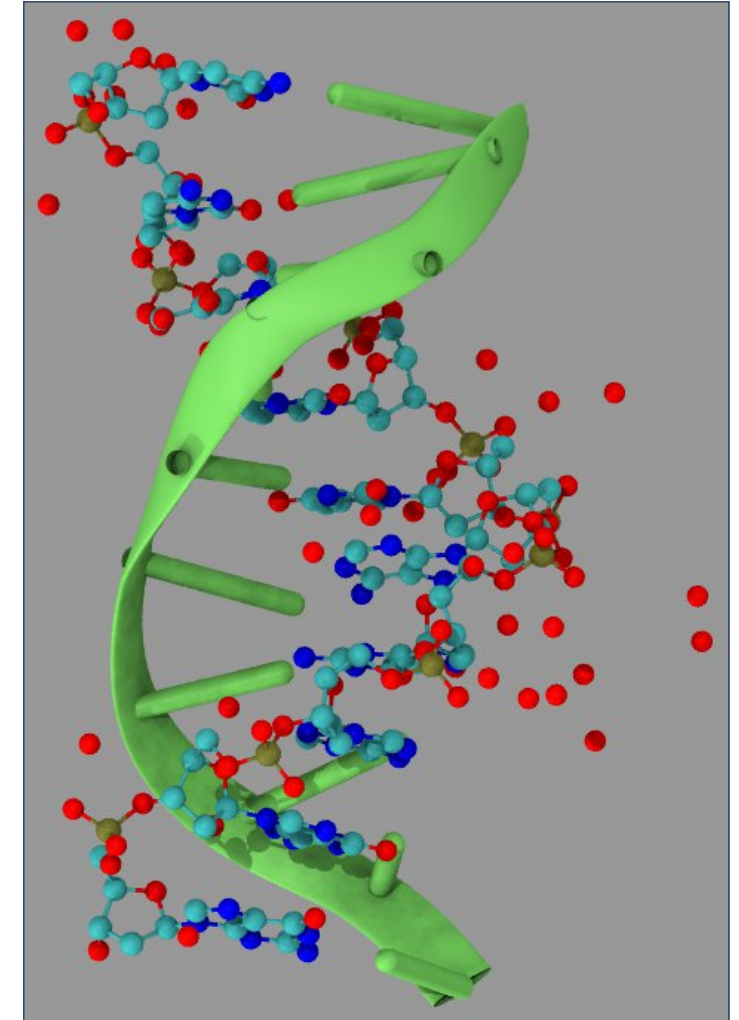


VMD: Summary

- Visualizes molecular structures (proteins, DNA, membranes)
- Displays MD trajectories
- Supports multiple formats: **PDB, PSF, DCD, GRO, XTC, etc**
- Advanced rendering: cartoon, licorice, surface, VDW
- Plugins for analysis (RMSD, hydrogen bonds)

Advanced:

- Supports **Tcl and Python scripting** for automation
- <http://www.ks.uiuc.edu/Training/Tutorials/#vmd>



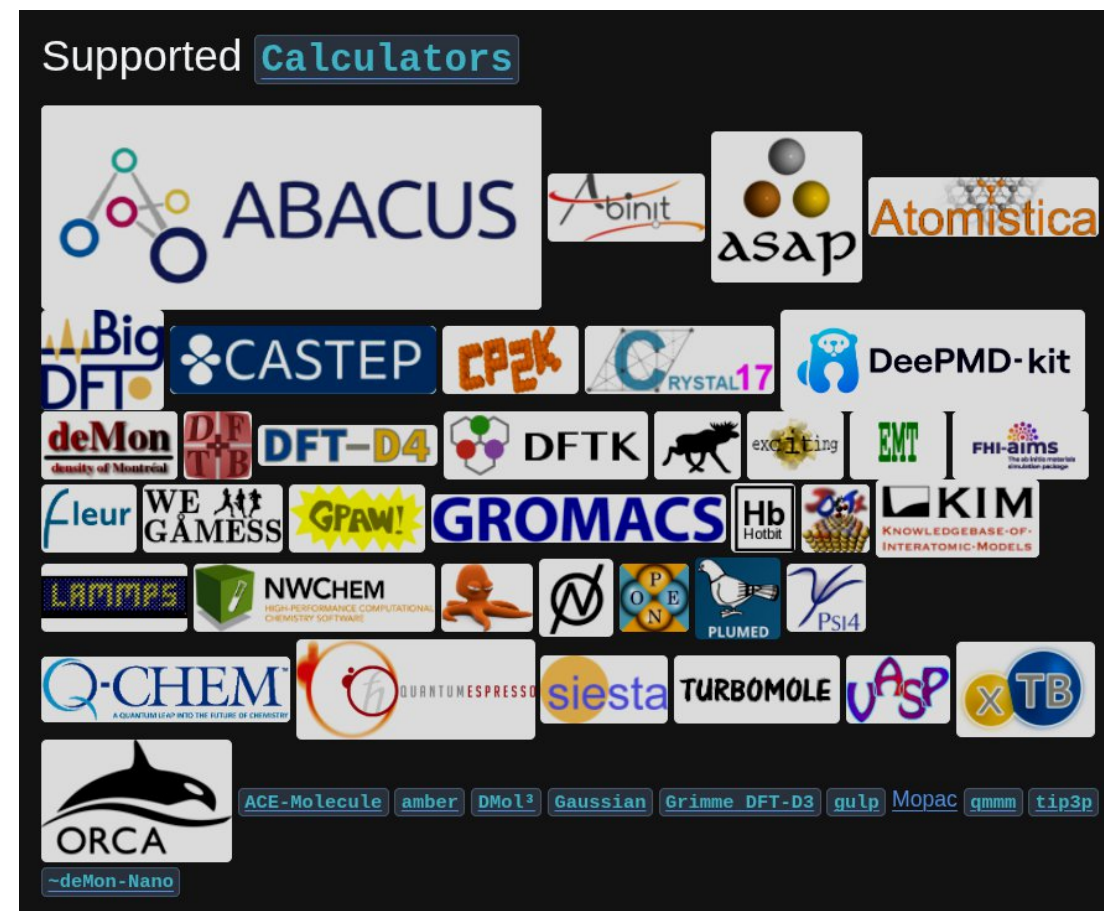
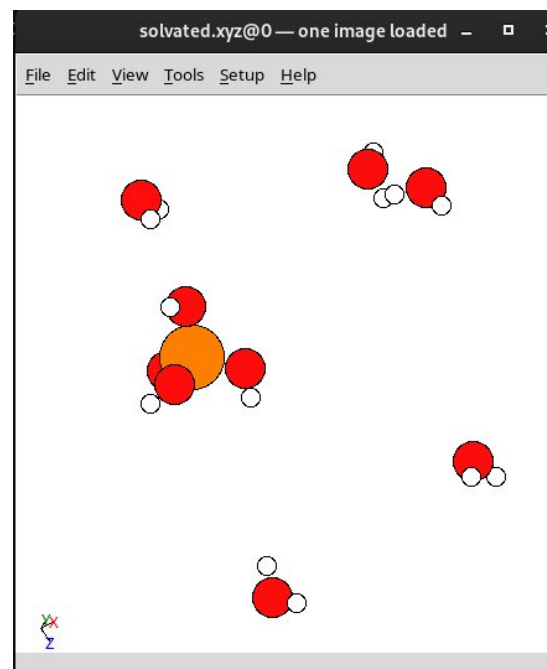
ATOMIC SIMULATION ENVIRONMENT - ASE

Introduction to Atomic Simulation Environment (ASE)



- Setting up, steering and analyzing: Python based
- Supports many Calculators/Software
- Easy to use (GUI, CLI)
- Open-source and cross-platform
- Data Analysis tools
- Movie making and Images

```
$ ase --help
$ ase sub-command --help
$ python -m module --help
```



Go to folder:

BioPhyCourse-latest/Day2/03_CP2K_QMMM/data/01-Intro-ASE

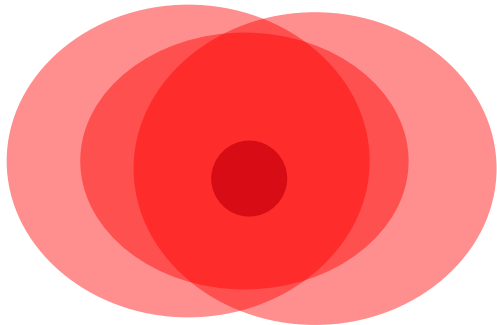
Open file:

MMS_ase_intro.ipynb

DENSITY FUNCTIONAL THEORY (DFT)

Time vs Length Scale

- hierarchy of theories available
- with accuracy computational cost increases
- limits system size and simulation duration
- a wide range of approximations



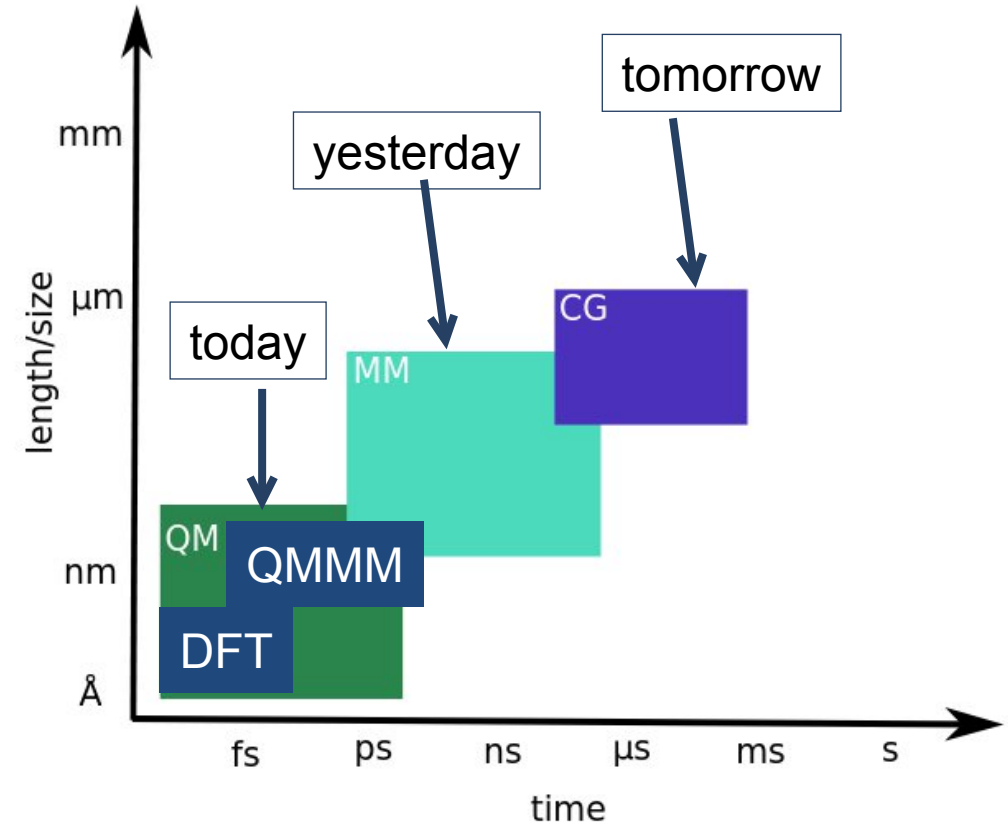
QM: Nuclei + Electrons



MM: Point Charge

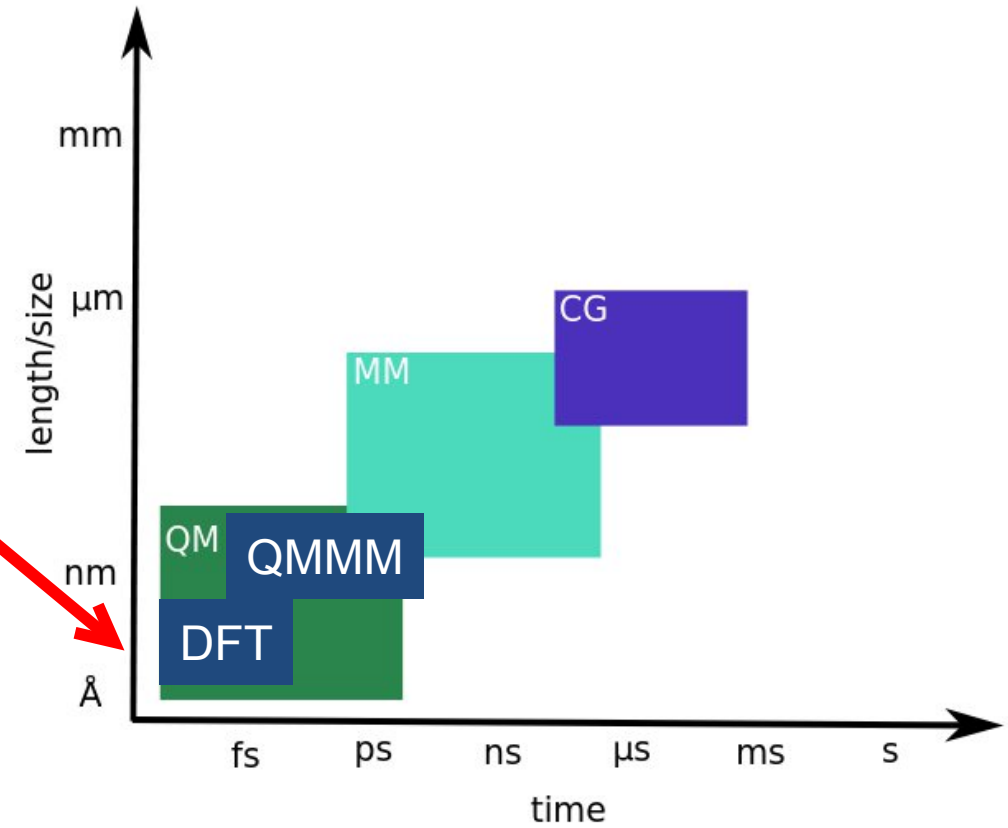


CG: Group of atoms - bead



Time vs Length. Adopted from [1]

DFT: Density Functional Theory

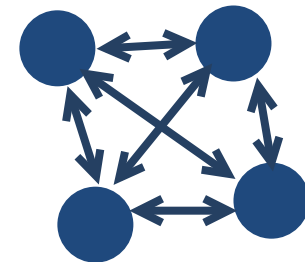


Time vs Length. Adopted from [1]

Density Functional Theory

DFT

- ground state electron density($\rho(\mathbf{r})$) alone determines the system behavior
- overcomes many-body problem



CP2K - Quickstep Method - DFT GPW [1]

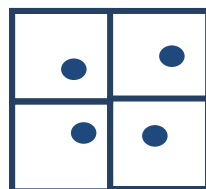
- Uses both Gaussian and Planewaves to speedup calculation
- **Gaussian**: Localized basis set centered on atomic positions – Suitable for molecules
- **Planewave**: Best suited for condensed matter, solids, periodic system

$$E[\rho(\mathbf{r})] = E^T[\chi(\mathbf{r})] + E^{\text{ne}}[\rho(\mathbf{r})] + E^{\text{ee}}[\rho(\mathbf{r})] + E^{\text{nn}} + E^{\text{XC}}[\rho(\mathbf{r})]$$

K.E electrons	nuclear-electron	electron-electron	nuclear-nuclear	XC
Gaussian	Gaussian+PW	PW	PW	PW

CP2K InputFile

$$E[\rho(\mathbf{r})] = E^T[\chi(\mathbf{r})] + E^{\text{ne}}[\rho(\mathbf{r})] + E^{\text{ee}}[\rho(\mathbf{r})] + E^{\text{nn}} + E^{\text{XC}}[\rho(\mathbf{r})]$$



Grids/Mesh

QM Method and Settings

Fine tune until consistent solution is reached

```
&FORCE_EVAL
  &DFT
    CHARGE 0
    &MGRID
      NGRIDS 4
      REL_CUTOFF 30
    &END MGRID
    &QS
      METHOD GPW
      EPS_DEFAULT 1.0E-12
      EXTRAPOLATION ASPC
    &END QS
    &SCF
      SCF_GUESS RESTART
      EPS_SCF 1.0E-8
      CHOLESKY INVERSE
      &DIAGONALIZATION
        ALGORITHM STANDARD
      &END DIAGONALIZATION
      &MIXING
        METHOD BROYDEN_MIXING
        NBROYDEN 16
      &END MIXING
    &END SCF
  &END FORCE_EVAL
```

```
&XC
  &XC_FUNCTIONAL PBE
&END XC_FUNCTIONAL
  &VDW_POTENTIAL
    POTENTIAL_TYPE PAIR_POTENTIAL
    &PAIR_POTENTIAL
      PARAMETER_FILE_NAME dftd3.dat
      TYPE DFTD3
      CALCULATE_C9_TERM .TRUE.
      REFERENCE_C9_TERM .TRUE.
      REFERENCE_FUNCTIONAL PBE
    &END PAIR_POTENTIAL
  &END VDW_POTENTIAL
&END XC
&END DFT
&END FORCE_EVAL
...
VdW and XC
```

```
&CELL      cell dimensions
  PERIODIC XYZ
  A 17.8    0.    0.
  B 0.      19.15 0.
  C 0.      0.    23.21
&END CELL
```

```
&KIND 0
  BASIS_SET SZV-MOLOPT-GTH
  POTENTIAL GTH-PBE
&END KIND
&KIND H
  BASIS_SET SZV-MOLOPT-GTH
  POTENTIAL GTH-PBE
&END KIND
Basis Sets and N
```

```
calc = CP2K(label='benzene',
            cutoff=250,
            basis_set='SZV-MOLOPT-GTH',
            pseudo_potential='auto',
            charge=0,
            xc='PBE',
            print_level='LOW',
            max_scf=20,
            inp=inp)
```

- project name
- electronic cutoff
- electronic wavefunction
- nuclear
- total System Charge
- exchange correlation
- output setting
- max number of SCF cycles

Example problem overview

Importance of Phosphorus

Essential for plant and animal life



Part of DNA, RNA



Brain functioning



Oxygen delivery



Photosynthesis [1]



Improve flowering



Root development

Limiting nutrient for plant growth

- Only small amount of P in soil
- Unlike C, N, O, H: P no gaseous phase
- Forests evolved strategies [2]
- Used in fertilizers: global food security



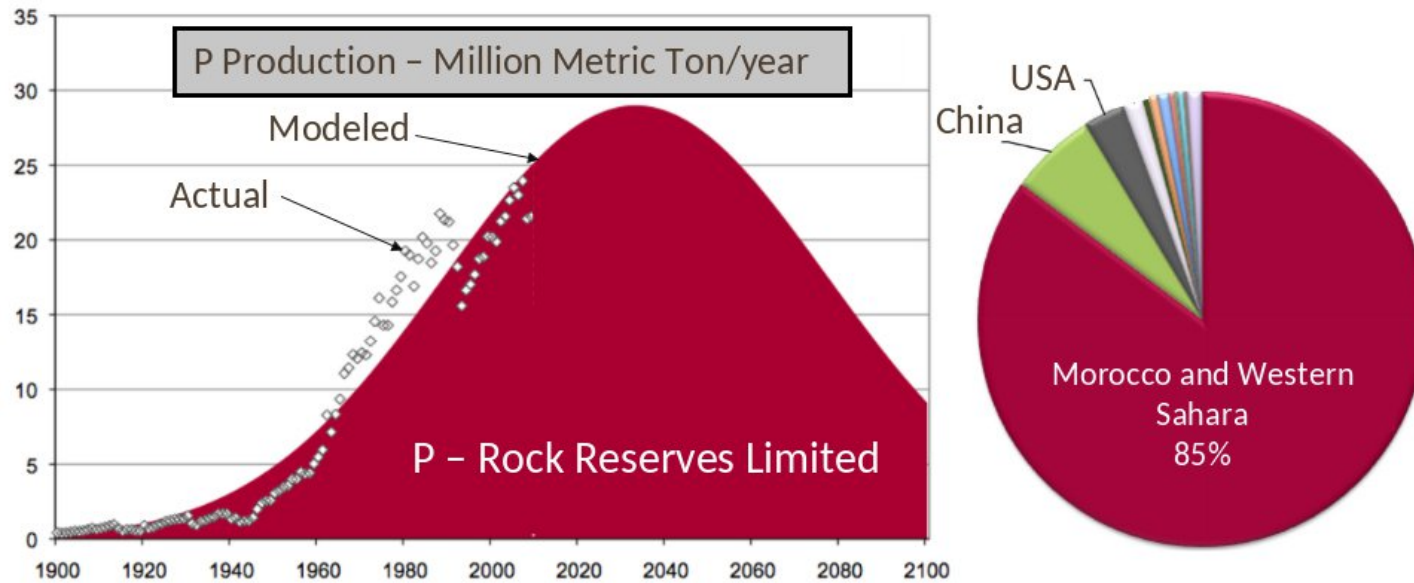
[1] Plants 12(15), 2861 (2023)

[2] Plant and Soil 348, 29 (2011)

Example problem overview

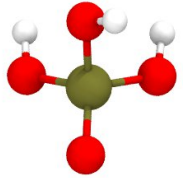
P Peak Scenario

- Peak P scenario could happen soon [1]
- Oil Crisis in 1970; NO alternative for P



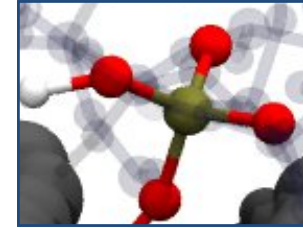
[1] Sustainability, 3(10), 2027-2049 (2011)

Example problem overview



Phosphoric acid (H_3PO_4)

solvated



Phosphate (HPO_4^{-2}) [2]

[2] Soil Sci. Plant Anal. 11, 835–842 (1980)

AIM

BioPhyCourse-latest/Day2/03_CP2K_QMMM/data/02-CP2K-DFT

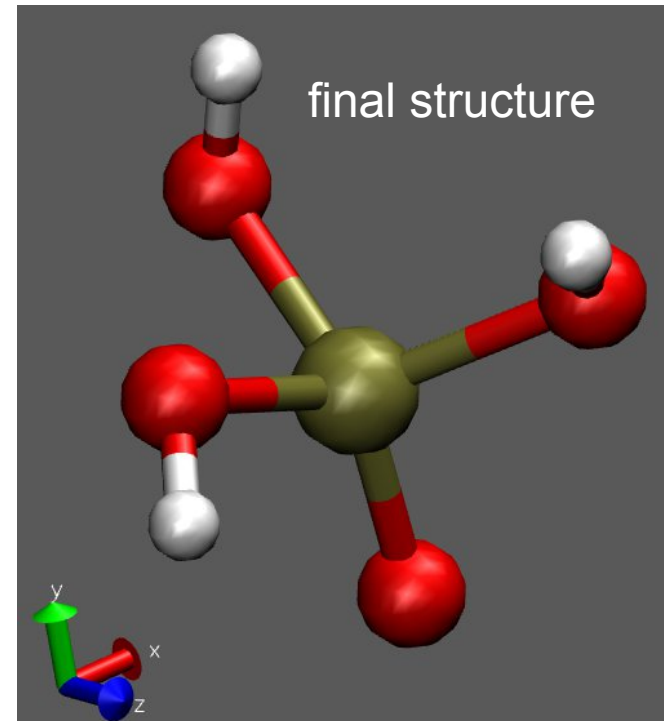
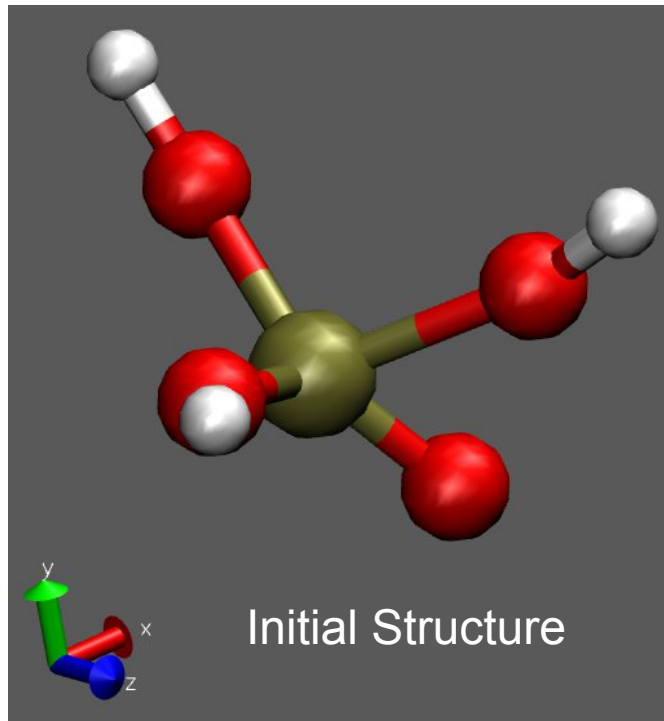
01-CP2K-Intro: Here we optimize Phosphoric acid using CP2K inputfile

02-CP2K-ASE: Here we solvate Phosphoric acid to see if it deprotonates using ASE+CP2K

Example problem overview

BioPhyCourse-latest/Day2/03_CP2K_QMMM/data/02-CP2K-DFT/01-CP2K-Intro/:

Here we optimize Phosphoric acid using CP2K inputfile

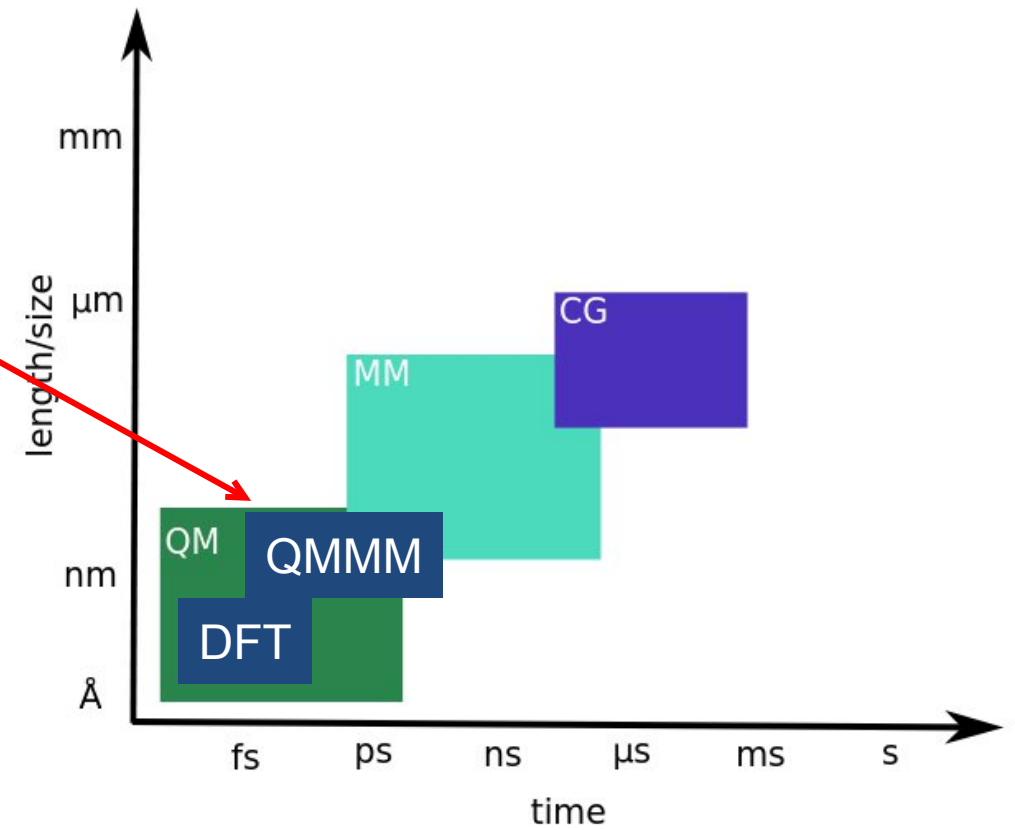


Example problem overview

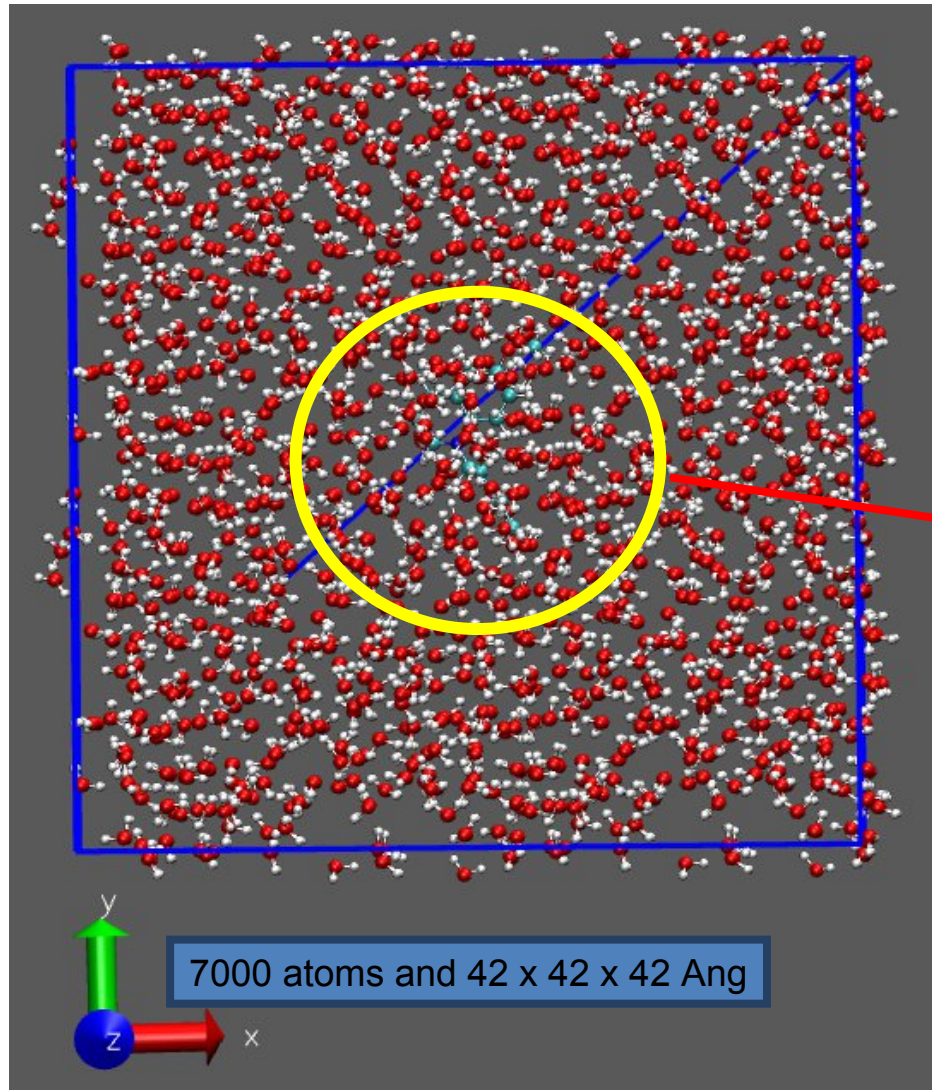
BioPhyCourse-latest/Day2/03_CP2K_QMMM/data/02-CP2K-DFT:

Here we solvate Phosphoric acid to see if it deprotonates using ASE+CP2K

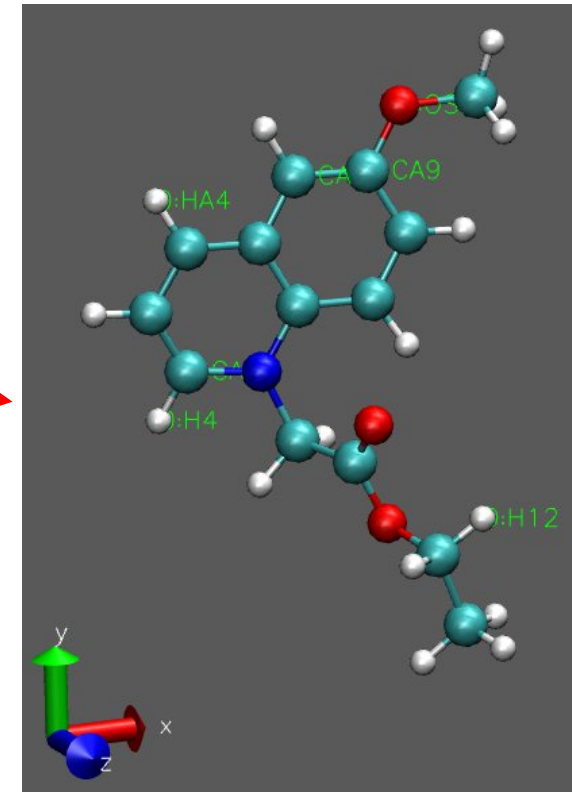
QM/MM



Why QM/MM

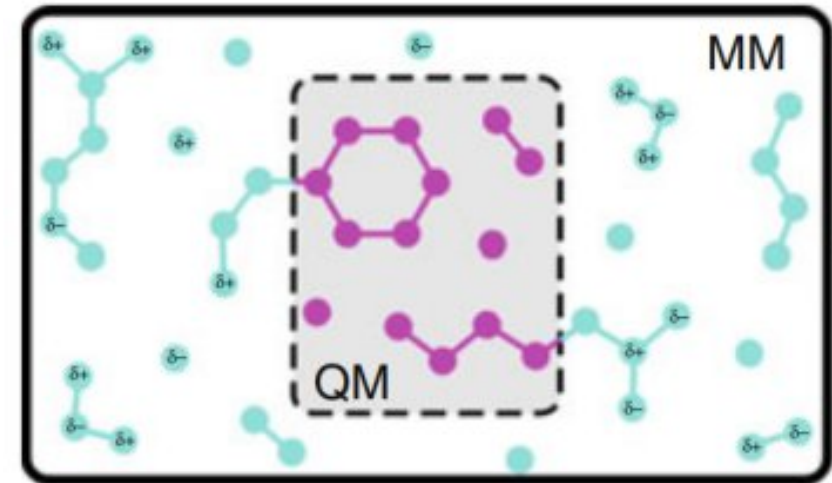
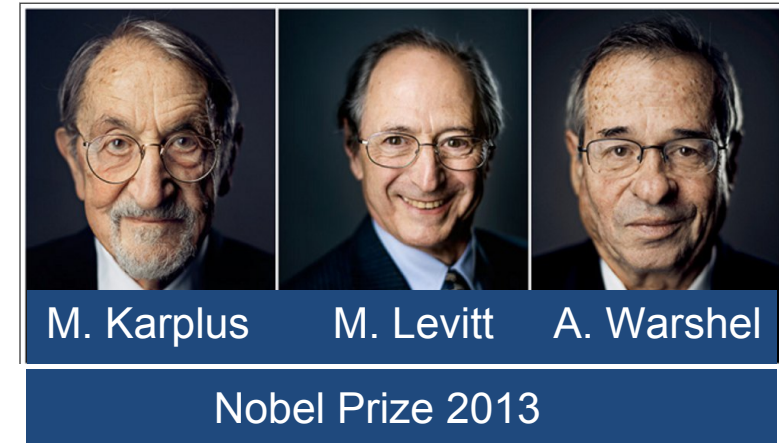


N-(6-methoxyquinolyl) acetoethyl ester



Why QM/MM

- Pure QM: expensive for 1000+ atoms
- Pure MM: not all reactions defined
- Relevant part with QM: Density Functional Theory [1]
- Rest with low level MM: Mostly All - Atoms FF [2,3]
- QM/MM Coupling [4,5]
- Advantage: Longer & accurate simulations at reduced cost [5]



[1] J. Chem. Phys. 152, 194103 (2020)

[2] J. Phys. Chem. B 108, 1255–1266 (2004)

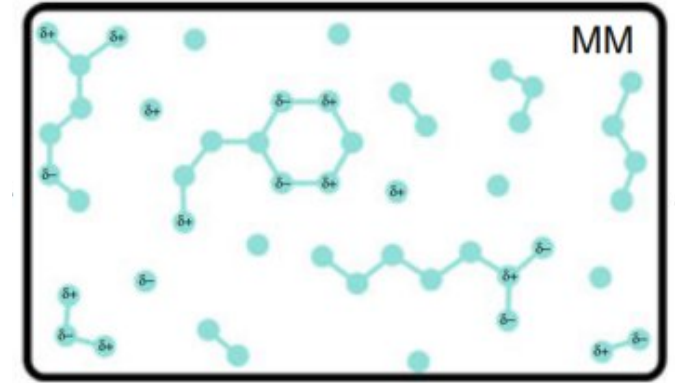
[3] J. Phys. Chem. 91, 6269–6271 (1987)

[4] J. Chem. Theory Comput. 1, 1176–1184 (2005)

[5] Angew. Chem. 48, 1198–1229 (2009)

Molecular Mechanics (MM)

- Functional forms and parameters define potential energy
- potentials derived from experiments or QM level simulations
- Electrons neglected in the model
- Cannot explain bond-breaking or forming, charge transfer, polarization
- need to parameterize of each system



$$E_{\text{system}} = \underbrace{\sum k_b(r_{ij} - r_0)^2 + \sum k_\theta(\theta_{ijk} - \theta_0)^2 + \sum k_\phi[1 + \cos(n\phi - \delta)]}_{E_{\text{bonded}}} + \underbrace{\sum_{\substack{i,j \\ i \neq j}} e^2 \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}} + \sum_{\substack{i,j \\ i \neq j}} \epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - 2 \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right]}_{E_{\text{nonbonded}}}$$

$$E_{\text{system}} = E_{\text{bonded}} + E_{\text{nonbonded}}$$

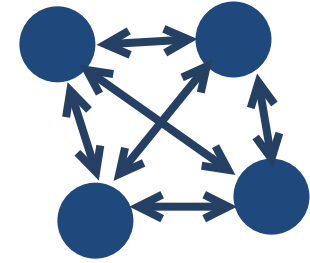
$$\sigma_{ij} = \frac{1}{2}(\sigma_i + \sigma_j)$$

$$\epsilon_{ij} = \sqrt{\epsilon_i \epsilon_j}$$

Density Functional Theory (QM)

DFT

- ground state electron density($\rho(\mathbf{r})$) alone determines the system behavior
- overcomes many-body problem



Quickstep Method - DFT GPW [1]

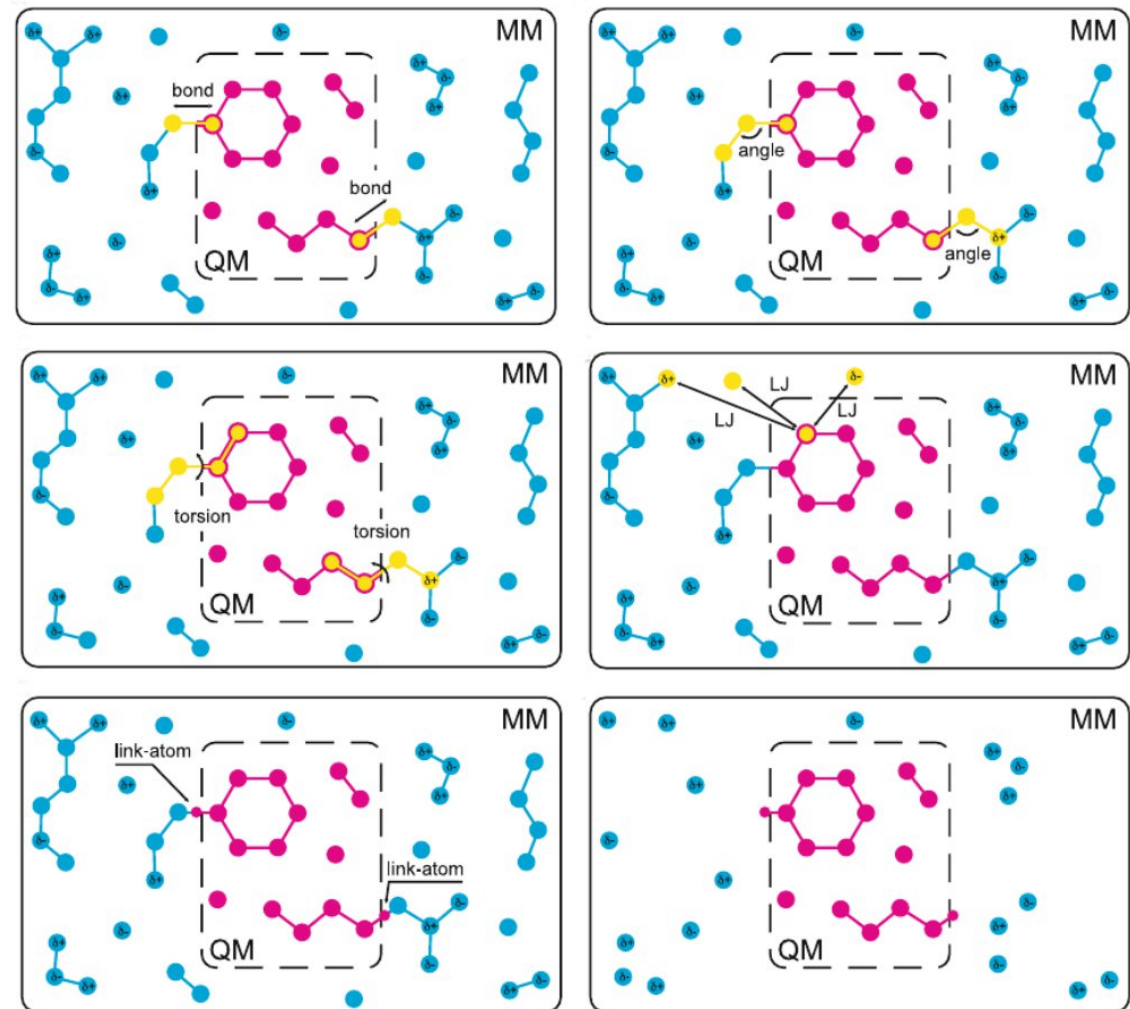
- Uses both Gaussian and Planewaves to speedup calculation
- **Gaussian**: Localized basis set centered on atomic positions – Suitable for molecules
- **Planewave**: Best suited for condensed matter, solids, periodic system

$$E[\rho(\mathbf{r})] = E^T[\chi(\mathbf{r})] + E^{\text{ne}}[\rho(\mathbf{r})] + E^{\text{ee}}[\rho(\mathbf{r})] + E^{\text{nn}} + E^{\text{XC}}[\rho(\mathbf{r})]$$

K.E electrons	nuclear-electron	electron-electron	nuclear-nuclear	XC
Gaussian	Gaussian+PW	PW	PW	PW

Additive QM/MM

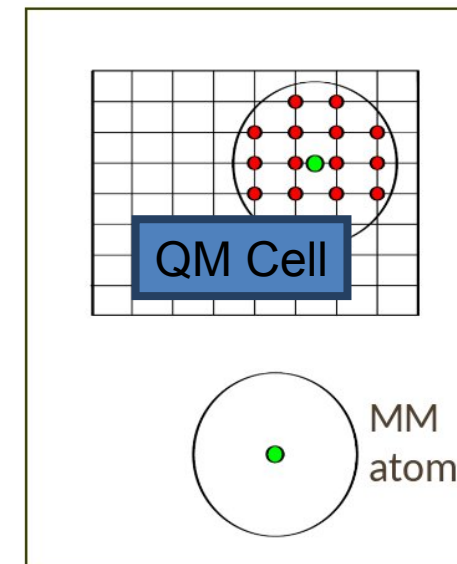
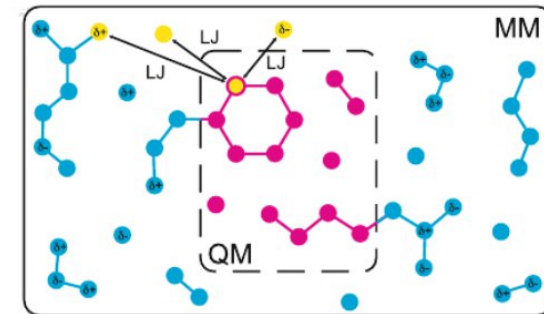
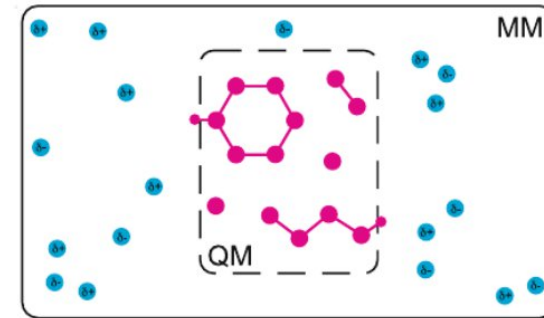
- $E(\mathbf{r}_\alpha, \mathbf{r}_a) = E^{QM}(\mathbf{r}_\alpha) + E^{MM}(\mathbf{r}_a) + E^{QM/MM}(\mathbf{r}_\alpha, \mathbf{r}_a)$
- Mechanical embedding: Electronic wavefunction isolated
 - QM-MM interactions defined at MM level
- Electrostatic Embedding: MM charges as one electron operators
 - MM polarize QM, but not vice versa
- Polarization Embedding: QM $\leftrightarrow$ MM polarization
 - Computationally expensive



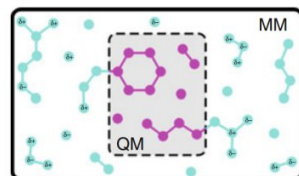
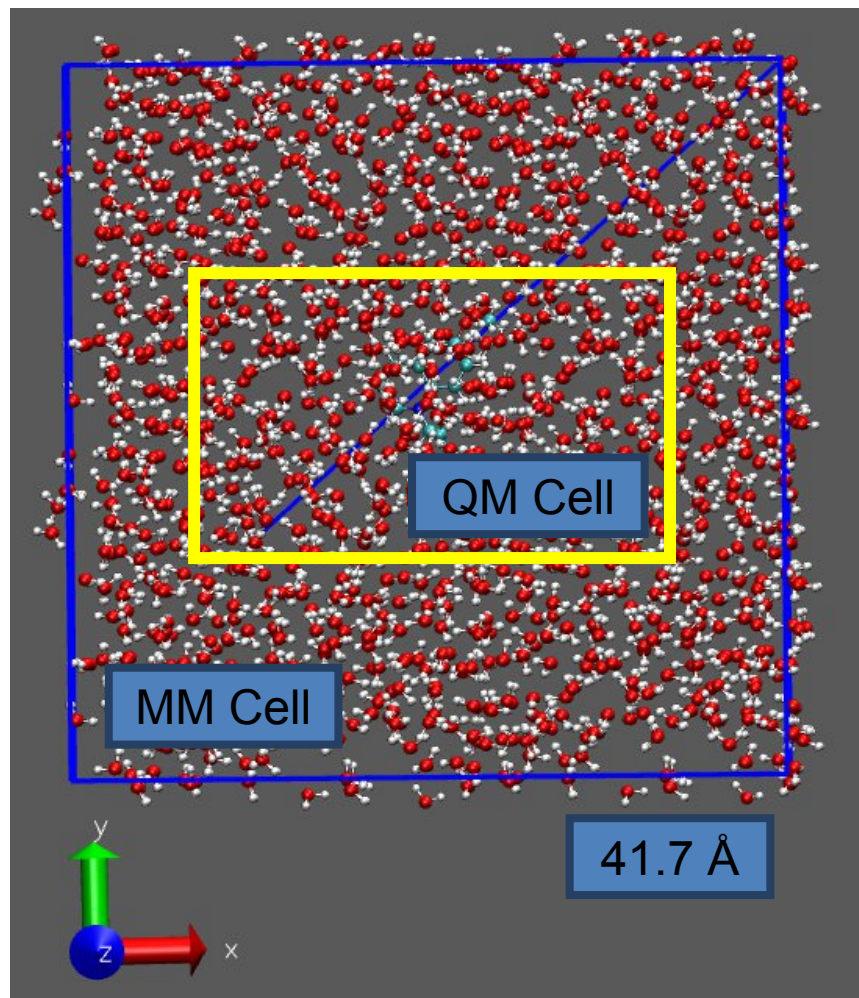
Electrostatic Embedding

- $E_{\text{NB}}^{\text{QM/MM}}(\mathbf{r}_i, \mathbf{R}_I) = \frac{1}{4\pi\epsilon_0} \sum_{i \in \text{MM}} q_i \int \frac{\rho}{|\mathbf{r} - \mathbf{r}_i|} d\mathbf{r} + \sum_{\substack{i \in \text{MM} \\ I \in \text{QM}}} \nu_{\text{vdW}}(\mathbf{r}_i, \mathbf{R}_I)$
charge density
electron + nuclei
↙
- MM atoms polarize the QM system
- charges of MM atoms and LJ parameters need to be parameterized
- electron-spill: electrons of QM atom close to MM atom gets trapped
- electrostatic embedding: GEEP
- MM Charge density in Gaussian

$$\frac{1}{(\sqrt{\pi} r_{c,a})^3} \exp\left(-\left(\frac{|\mathbf{r} - \mathbf{r}_i|}{r_{c,a}}\right)^2\right)$$



Example1: Overview



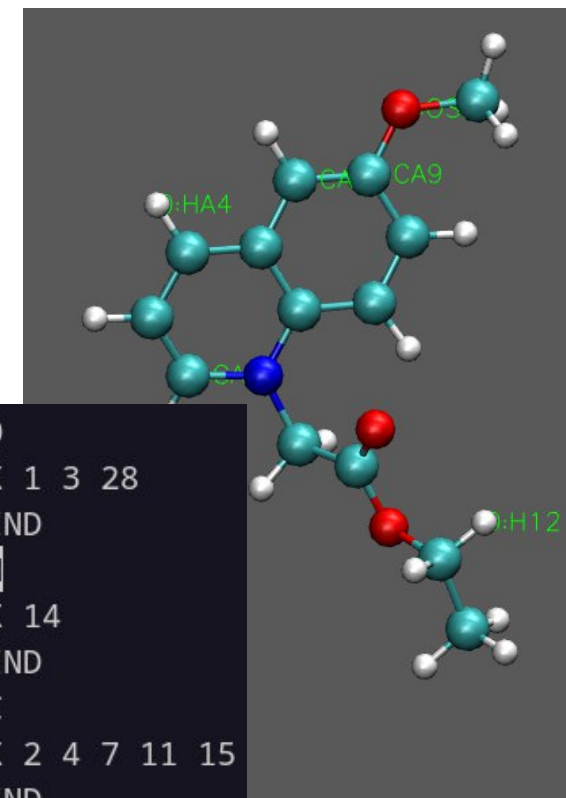
```
&QMMM ! This defines the QS cell i
ECOUPL GAUSS
USE_GEEP_LIB 15
&CELL
  ABC 13.79565 17.3825 10.76061
  PERIODIC XYZ
&END CELL
&PERIODIC # apply periodic poten
```

```
&SUBSYS
&CELL
  ABC 41.751 41.751 41.751
  ALPHA_BETA_GAMMA 90 90 90
  PERIODIC XYZ
&END CELL
```

```
&KIND H
  BASIS_SET DZVP-MOLOPT-GTH
  ELEMENT H
  POTENTIAL GTH-BLYP-q1
&END KIND
&KIND C
  BASIS_SET DZVP-MOLOPT-GTH
  ELEMENT C
  POTENTIAL GTH-BLYP-q4
&END KIND
```

ATOM	8	HC	17.593	24.836	20.763
ATOM	9	HC	19.204	25.485	21.124
ATOM	10	HC	18.086	25.028	22.422
ATOM	11	C32	20.744	21.568	18.721
ATOM	12	H13	21.474	22.379	18.609
ATOM	13	H13	19.956	21.768	18.003
ATOM	14	NA	21.342	20.240	18.435
ATOM	15	CA8	22.670	20.212	17.952

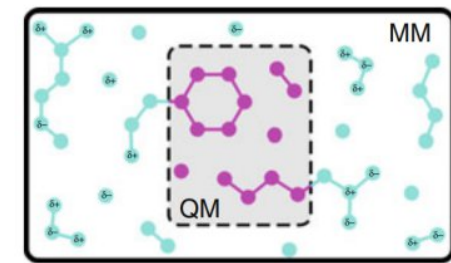
N-(6-methoxyquinolyl) acetoethyl ester



```
&QM_KIND O
  MM_INDEX 1 3 28
&END QM_KIND
&QM_KIND N
  MM_INDEX 14
&END QM_KIND
&QM_KIND C
  MM_INDEX 2 4 7 11 15
&END QM_KIND
&QM_KIND H
```

QMMM Inputfile

BioPhyCourse-latest/Day2/03_CP2K_QMMM/data/03-CP2K-QMMM/QMMM_MQAE



```
&FORCE_EVAL
  METHOD QMMM
  &DFT
    BASIS_SET_FILE_NAME BASIS_MOLOPT
    CHARGE 1
    MULTIPLICITY 1
    POTENTIAL_FILE_NAME POTENTIAL
  &MGRID
    COMMENSURATE
    CUTOFF 400
  &END MGRID
  &QS
    EPS_DEFAULT 1.0E-12
    METHOD GPW
  &END QS
  &SCF
    EPS_SCF 1.0E-6
    MAX_SCF 300
  &OT T
    MINIMIZER DIIS
    PRECONDITIONER FULL_ALL
    STEPSIZE 1.4999999999999999E-01
  &END OT
  &END SCF
  &XC
    &XC_FUNCTIONAL BLYP
    &END XC_FUNCTIONAL
  &END XC
  &END DFT
  &MM
    &FORCEFIELD
```

DFT Section

```
&END DFT
&MM
  &FORCEFIELD
    EI_SCALE14 1.0
    PARMTYPE AMBER
    PARM_FILE_NAME MQAE.prmtop
    VDW_SCALE14 1.0
  &SPLINE
    EMAX_SPLINE 1.0E14
    RCUT_NB [angstrom] 12
  &END SPLINE
  &END FORCEFIELD
  &POISSON
    &EWALD
      ALPHA .40
      EWALD_TYPE SPME
      GMAX 80
    &END EWALD
  &END POISSON
  &END MM
  &QMMM ! This defines the QS cell in t
```

MM Section

```
&END POISSON
&END MM
&QMMM ! This defines the QS cell i
  ECOUPL GAUSS
  USE_GEEP_LIB 15
  &CELL
    ABC 13.79565 17.3825 10.76061
    PERIODIC XYZ
  &END CELL
  &PERIODIC # apply periodic poten
    #turn on/off coupling/recouplin
  &MULTIPOLE ON
  &END MULTIPOLE
  &END PERIODIC
  &QM_KIND O
    MM_INDEX 1 3 28
  &END QM_KIND
  &QM_KIND N
    MM_INDEX 14
  &END QM_KIND
  &QM_KIND C
    MM_INDEX 2 4 7 11 15 17 19 21 2
  &END QM_KIND
  &QM_KIND H
    MM_INDEX 5 6 8 9 10 12 13 16 18
  &END QM_KIND
  &END QMMM
  &SUBSYS
    &CELL
      ABC 41.751 41.751 41.751
```

QMMM

QM Cell

QM
atoms

```
&END QMMM
&SUBSYS
  &CELL
    ABC 41.751 41.751 41.751
    ALPHA_BETA_GAMMA 90 90 90
    PERIODIC XYZ
  &END CELL
  &KIND H
    BASIS_SET DZVP-MOLOPT-GTH
    ELEMENT H
    POTENTIAL GTH-BLYP-q1
  &END KIND
  &KIND C
```

Total Cell

BASIS SET

```
  &ELEMENT H
    POTENTIAL GTH-BLYP-q5
  &END KIND
  &TOPOLOGY
    CONN_FILE_FORMAT AMBER
    CONN_FILE_NAME MQAE.prmtop
    COORD_FILE_FORMAT PDB
    COORD_FILE_NAME MQAE.pdb
  &END TOPOLOGY
  &END SUBSYS
&END FORCE_EVAL
```

FF Data

Topology

CONN_FILE_FORMAT: *enum* = **GENERATE**

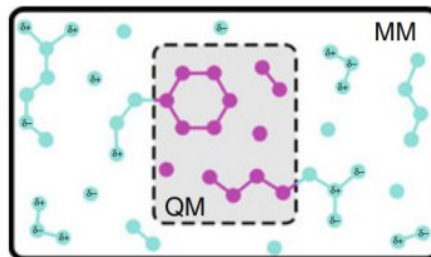
Aliases: CONNECTIVITY

Usage: **CONN_FILE_FORMAT** (PSF|UPSF|MOL_SET|GENERATE|OFF|G87|G96|AMBER|USER)

Valid values:

- **PSF** Use a PSF file to determine the connectivity. (support standard CHARMM/XPLOR and EXT CHARMM)
- **UPSF** Read a PSF file in an unformatted way (useful for not so standard PSF).
- **MOL_SET** Use multiple PSF (for now...) files to generate the whole system.
- **GENERATE** Use a simple distance criteria. (Look at keyword BONDPARM)
- **OFF** Do not generate molecules. (e.g. for QS or ill defined systems)
- **G87** Use GROMOS G87 topology file.
- **G96** Use GROMOS G96 topology file.
- **AMBER** Use AMBER topology file for reading connectivity (compatible starting from AMBER V.7)
- **USER** Allows the definition of molecules and residues based on the 5th and 6th column of the COORD section. This option can be handy for the definition of molecules with QS or to save memory in the case of very large systems (use PARA_RES off).

Ways to determine and generate a molecules. Default is to use GENERATE [\[Edit on GitHub\]](#)



COORD_FILE_FORMAT: *enum* = **OFF**

Aliases: COORDINATE

Usage: **COORD_FILE_FORMAT** (OFF|PDB|XYZ|G96|CRD|CIF|XTL|CP2K)

Valid values:

- **OFF** Coordinates read in the &COORD section of the input file
- **PDB** Coordinates provided through a PDB file format
- **XYZ** Coordinates provided through an XYZ file format
- **G96** Coordinates provided through a GROMOS96 file format
- **CRD** Coordinates provided through an AMBER file format
- **CIF** Coordinates provided through a CIF (Crystallographic Information File) file format
- **XTL** Coordinates provided through a XTL (MSI native) file format
- **CP2K** Read the coordinates in CP2K &COORD section format from an external file. NOTE: This file will be overwritten with the latest coordinates.

Set up the way in which coordinates will be read. [\[Edit on GitHub\]](#)

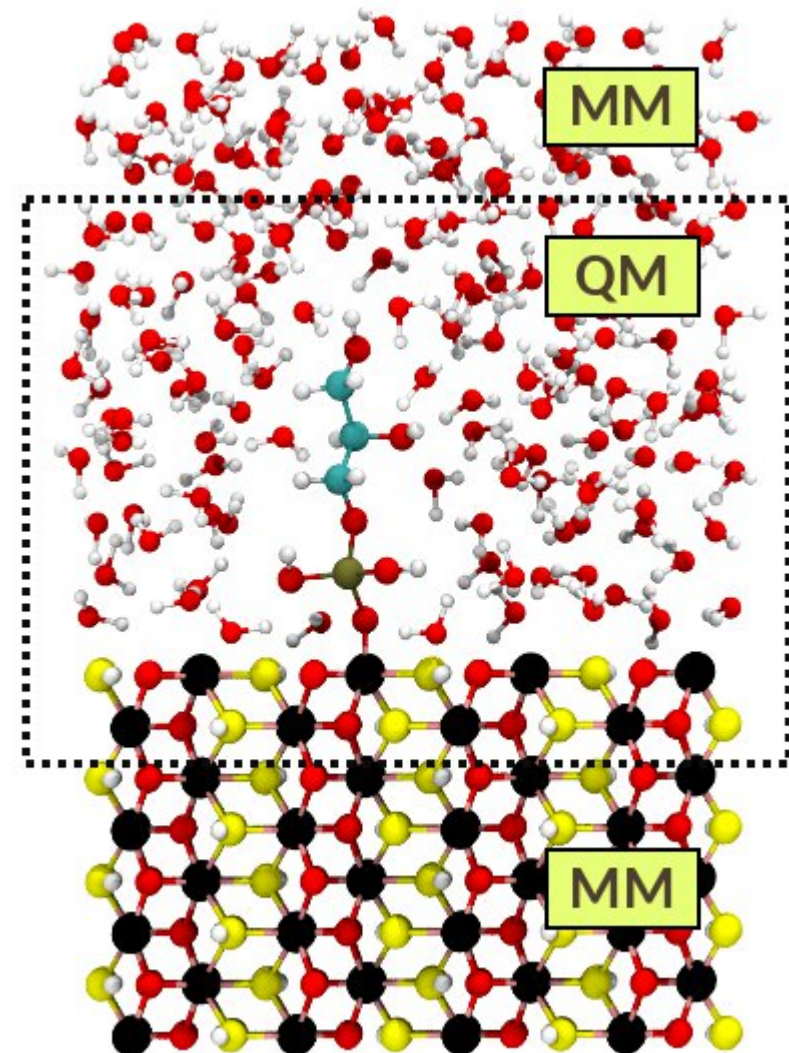
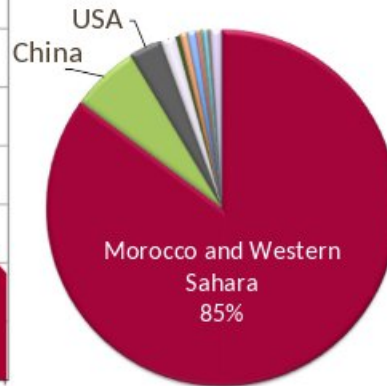
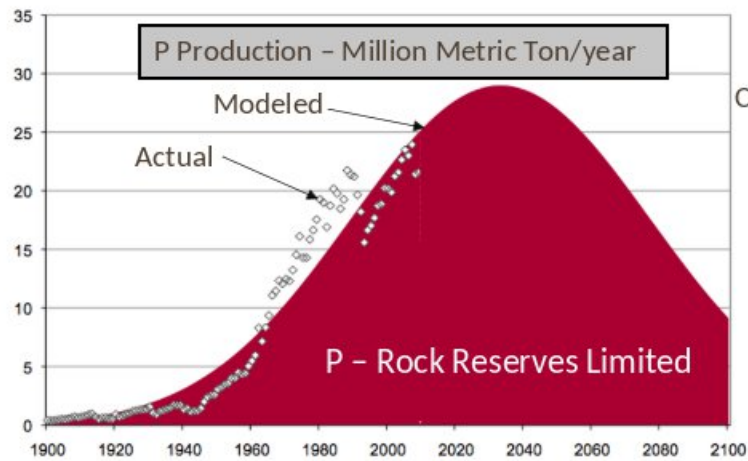
```
!
POTENTIAL GTH-BLYP-q5
&END KIND
&TOPOLOGY
  CONN_FILE_FORMAT AMBER
  CONN_FILE_NAME MQAE.prmtop
  COORD_FILE_FORMAT PDB
  COORD_FILE_NAME MQAE.pdb
&END TOPOLOGY
&END SUBSYS
&END FORCE EVAL
```

Inputfile

Example 2

P Peak Scenario

- Peak P scenario could happen soon
- Oil Crisis in 1970; NO alternative for P

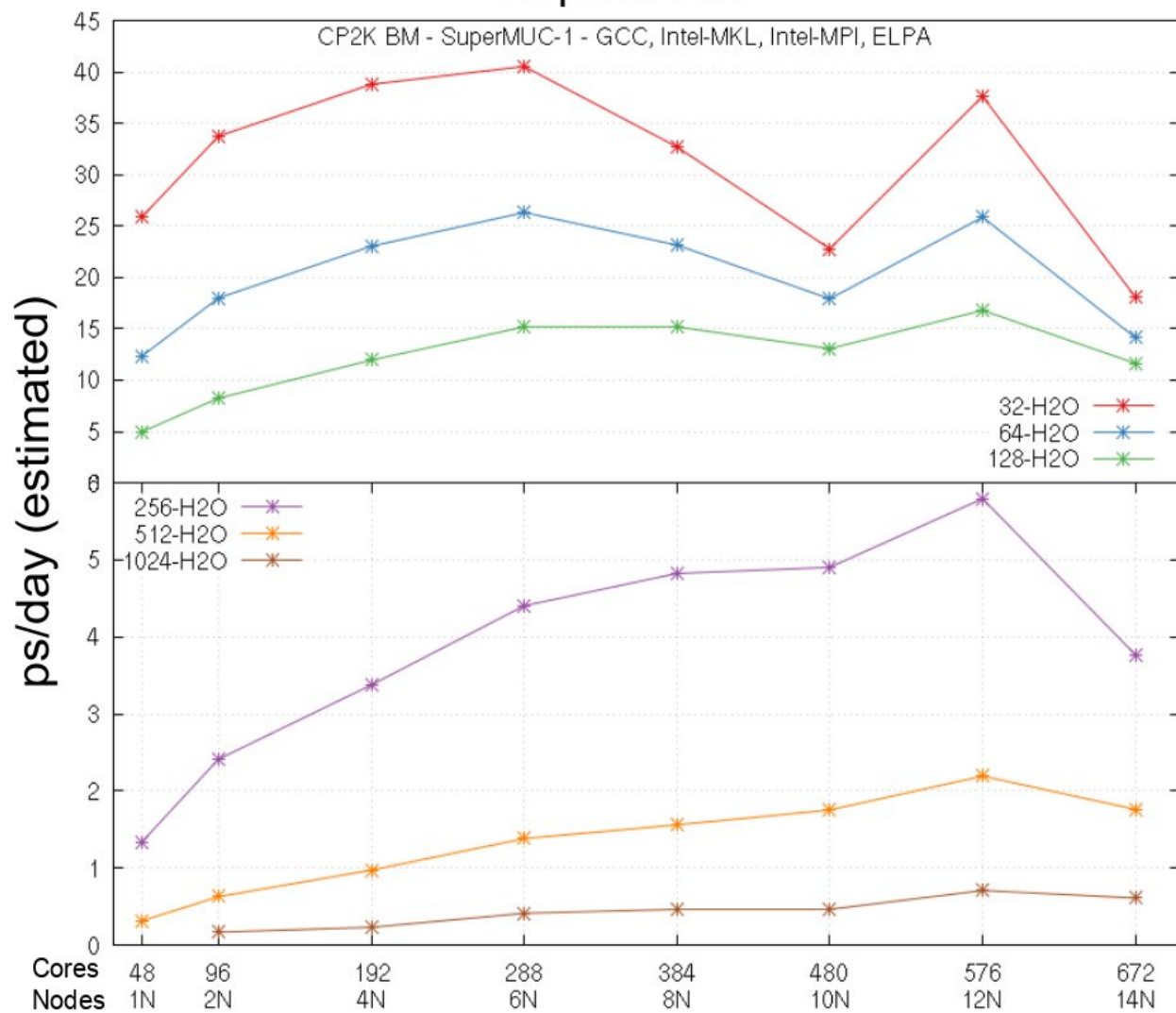


Benchmarks

1picosecond simulation of
1000 water molecules (3000 atoms
in 1 and half day.



SuperMUC1



SuperMUC2

